
Design and Optimisation of Mammalian Cell Perfusion Cultures

In this chapter, we discuss state-of-the-art strategies for perfusion process design, development and optimisation. We first introduce the essential steps and boundaries for the development of a perfusion process. Next, we evaluate clone and media screening for perfusion processes and compare various alternative scale-down systems. Once the expression system is selected, we proceed with the design of the reactor operating conditions to maximise volumetric productivity and yield while reducing medium consumption. The fundamental issue of product quality attributes for biologics and biosimilars and the impact of perfusion operating mode are also discussed. Finally, we introduce the main objectives in the scale-up of the developed process to clinical and commercial productions.

4.1 PRINCIPLES OF OPTIMAL PERFUSION BIOREACTOR DESIGN

The implementation of a perfusion process has to address some issues, which refer not only to production costs and product quality but also to process robustness. The most reasonable starting point is probably to select the maximum volumetric throughput while maximising the protein-producing biomass inside the bioreactor, in order to improve process economics. In Chapter 2, we have seen that a key parameter in this regard is the ratio of volumetric throughput, represented by the perfusion rate, P (RV/day), and the viable cell density, X_V (10^6 cells/mL), defined as the cell-specific perfusion rate, CSPR (pL/cell/day). The CSPR defines the amount of medium fed to a single cell per day, and therefore, reducing the CSPR represents an important factor in optimal process design (Konstantinov et al. 2006).

In the following, we discuss the robust design of perfusion cultures going through, first, the proper selection of clones and media; next, the determination of a suitably low CSPR, which in the best case is equal to the $\text{CSPR}_{\min}$ value defined in Section 4.3; and, finally, the best operating conditions in terms of viable cell density and perfusion rate at the given $\text{CSPR}_{\min}$.

4.1.1 The critical Bioreactor Parameters

In Chapter 2, we discussed the strategies to control the system variables – such as pH, DO, agitation, working volume, flowrates of the inlet and outlet streams, and viable cell density. The implementation of such strategies requires the definition of proper

upper and lower boundaries, which are particularly relevant for viable cell density and perfusion rate or CSPR.

In the case of the perfusion rate, the upper limit is mainly determined by two factors: set-up operation sustainability and economic convenience. Indeed, the cell retention device cannot sustain too large perfusion rates and therefore imposes limitations, while, on the other hand, high perfusion rates imply high medium consumption and consequently high operating costs. A value often considered for process development is a perfusion rate of one reactor volume per day (RV/day), but depending on the specific characteristics of the system under consideration, higher perfusion rates may also be favourable. The lower bound of the perfusion rate, on the other hand, depends on multiple factors, including the cell-specific growth rate, the medium formulation and cost, and the stability of the product. Low perfusion rates lead to long product residence times in the reactor favouring post-translational modifications of the product with negative consequences on product quality attributes. For a given cell density, as the perfusion rate is decreased, metabolite levels can be reached, below which further cultivation is not possible due to induced cell death, decline in cell-specific productivity or decline in product quality (Konstantinov et al. 2006).

In the case of viable cell density, X_V , it is obvious that, in principle, larger values are always to be preferred so that only upper limits actually need to be considered. In addition to the minimum cell-specific perfusion rate, which – at a given medium throughput – constrains the maximum viable cell density, other limitations come from oxygenation, viscosity of the cell culture broth and cell retention. As mentioned before, running the cell culture below the minimum CSPR does not allow a stable operation; thus, for a given perfusion rate, the maximum VCD is constrained by the depth of the used medium, which defines the CSPR_{min} value (Konstantinov et al. 2006). On the other hand, perfusion cultures are typically operated at much higher viable cell densities compared to batch cultivations, which implies larger oxygen consumption and carbon dioxide production rates. These typically become a problem when targeting cell densities above about 100×10^6 cells/mL. Another limitation related to high cell density values is that the viscosity increases, which results in higher stirring and agitation rates in order to guarantee complete homogenisation of the cell culture. This can lead to shear stress values that are harmful for the cells (Ozturk 1996). Finally, studies on filtration-based cell retention devices have shown that, depending on the specific retention technology adopted, limitations on the cell density value can occur. These limitations can be related to the residence time of the cells within the uncontrolled environment of the retention device; the characteristics of the device itself, such as the pump capacity with respect to tangential or alternating tangential flow filtration, or the porosity structure of the filtering media (Clincke et al. 2013b).

As discussed in Chapter 2, the concept of minimum cell-specific perfusion rate, CSPR_{min}, represents one of the most important control aspects in perfusion cultures. This is determined by the medium formulation and the expression system itself. Depending on cell line and clone, the nutrient requirements can greatly vary. Thus, early media and cell clone screening are prerequisites for process design for continuous cultures.

With respect to the expression system, for a given set of operating conditions (P, VCD, and CSPR), cellular metabolism for an ideal non-ageing cell should be fixed

4.1 Principles of Optimal Perfusion Bioreactor Design

and characterised by constant cell-specific productivity, q_P , and cell-specific growth rate, μ (Ozturk 2014):

$$q_P = \frac{g_{protein} \text{ produced}}{\text{cell} \times \text{time}} \quad \left[\frac{pg_{protein}}{\text{cell} \times \text{day}} \right] \quad (4.1)$$

$$\mu = \frac{\text{cell produced}}{\text{cell} \times \text{time}} \quad \left[\frac{1}{\text{day}} \right]. \quad (4.2)$$

These parameters are very important in defining the process performance, in terms of productivity, medium consumption and yield. Their value is defined by the cell metabolism, and therefore, it is a function of temperature, pH and composition of the supernatant, particularly in terms of nutrients and toxic metabolites.

It is worth noting that, ideally in a culture operated at constant viable cell-density, the cell specific productivity does not depend on the CSPR if the medium is deep enough (Konstantinov et al. 2006). This means that the CSPR can be decreased without changing q_P as long as no nutrient limitation is encountered or, for any other reason, the culture becomes unstable. This defines the CSPR_{min} value. On the other hand, the cell-specific growth rate, μ , decreases when decreasing the cell-specific perfusion rate, and therefore, at the CSPR_{min} the cell growth is minimum but still high enough to sustain a viable culture. These considerations highlight the relevance of the CSPR_{min} in determining the optimal operating conditions of the bioreactor.

4.1.2 The Optimal Bioreactor Design Procedure: Step I

In order to univocally define the bioreactor steady-state operation, we need to select values for the viable cell density, the perfusion rate and, consequently, the CSPR. For this, we consider three objectives, which define the performance of a given bioreactor: the volumetric productivity (PR), the yield (Y) and the medium consumption (MC), which are defined as follows:

$$PR = \frac{g_{protein} \text{ produced}}{V_t \times \text{time}} = (P - B)C_P \quad \left[\frac{g_{protein}}{L \times \text{day}} \right] \quad (4.3)$$

$$Y = \frac{(P - B)}{P} \quad (4.4)$$

$$MC = \frac{\text{media consumed}}{g_{protein} \text{ produced}} = \frac{P}{PR} \quad \left[\frac{L_{media}}{g_{protein}} \right], \quad (4.5)$$

where P represents the perfusion rate (RV/day), B is the bleed rate (RV/day), and C_P is the protein concentration inside the bioreactor or titre (g/L).

As seen in Section 2.6.1, the steady-state mass balances for cells and target protein lead to the following expressions for the bleed flowrate, B and the protein concentration, C_P , respectively:

$$B = \mu \quad (4.6)$$

$$C_P = \frac{q_P X_v}{P} = \frac{q_P}{\text{CSPR}}. \quad (4.7)$$

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Thus, substituting in Equations 4.2–4.4, the objective functions, PR , Y and MC at steady state can be written as follows:

$$PR = (P - \mu) \frac{q_p X_V}{P} \quad (4.8)$$

$$Y = \frac{(P - \mu)}{P} \quad (4.9)$$

$$MC = \frac{P^2}{(P - \mu) q_p X_V}. \quad (4.10)$$

When we consider the design of a perfusion process, we refer to the previously mentioned parameters to quantify the process performance, aside from any consideration about product quality. In particular, while the volumetric productivity and yield need to be maximised, the medium consumption should be minimised. As the protein removed in the bleed stream is not further processed, the bleed is considered as product loss and therefore, according to Equation (4.9), the cellular growth, μ should be minimised to maximise the process yield, which, as mentioned before, occurs at the $CSPR_{min}$. In general, reasonable values for the bleed stream and, therefore, for the process yield are considered in the range 5–10 per cent of the overall perfusion rate. When considering productivity and medium consumption, it is generally understood that operation at $CSPR_{min}$ gives optimal process performance (Konstantinov et al. 2006). In principle, two approaches can be envisioned to estimate the $CSPR_{min}$. The push-to-low approach, originally proposed by Konstantinov et al. (2006), consists of a stepwise reduction of the perfusion rate at a fixed viable cell density so as to explore decreasing values of the $CSPR$. Alternatively, the push-to-high approach involves a stepwise increase of the viable cell density at a constant perfusion rate. In both cases, the $CSPR$ decreases at each step until either medium limitations or metabolite inhibitions result in process instability (e.g., cell death or decrease in cell-specific productivity), which, in principle, occurs at the $CSPR$ value corresponding to the $CSPR_{min}$. The path followed to reach the $CSPR_{min}$ value in the approaches is not the same. In the first one, the perfusion rate decreases, which may lead to excessive average residence times in the case of fragile products, while in the second, the VCD increases, which may lead to unfeasibly large values. The two procedures are illustrated schematically in Figure 4.1.

If we consider the operation at constant perfusion rate, the optimisation procedure comprises the following step. We take a reasonable set of starting conditions for P and X_V and measure the volumetric productivity and medium consumption at steady state. As discussed in the following paragraphs, from Equations (4.8) and (4.10), we can see that by increasing the VCD, the push-to-high approach results in increased volumetric productivity and decreased medium consumption. This is continued until the culture remains stable – that is, until $CSPR$ becomes equal to $CSPR_{min}$, as indicated by the point in Figure 4.1. Accordingly, at this point, for the selected perfusion rate, the volumetric productivity is maximised and the medium consumption is minimised while providing a stable cultivation. This is illustrated for the operation at a constant perfusion rate of 1.0 RV/day in Figure 4.2.

Figure 4.2A describes the relation between VCD and perfusion rate for various values of $CSPR$. Considering a sequential bioreactor operation at constant perfusion rate of 1.0 RV/day and viable cell densities between 10 and 50×10^6 cells/mL,

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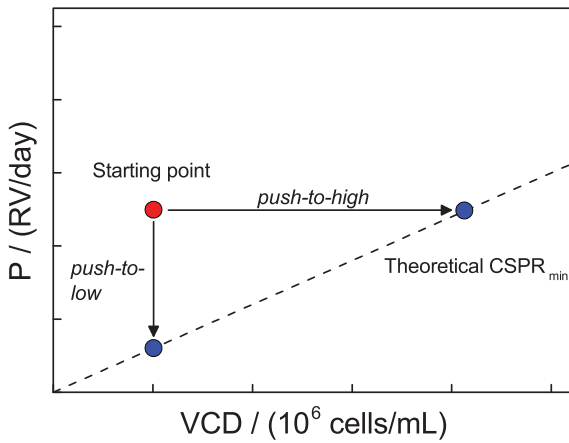


Figure 4.1 Schematic representation of the push-to-low and the push-to-high approach to determine $CSPR_{min}$. The red point represents the initial condition of the procedure and the slope of the broken straight lines the $CSPR$ value. © Reprinted from *Biochemical Engineering Journal*, 151, Moritz K. F. Wolf, Anna Pechlaner, Veronika Lorenz, Daniel J. Karst, Jonathan Souquet, Hervé Broly and Massimo Morbidelli, 'A two-step procedure for the design of perfusion bioreactors', 2019, with permission from Elsevier

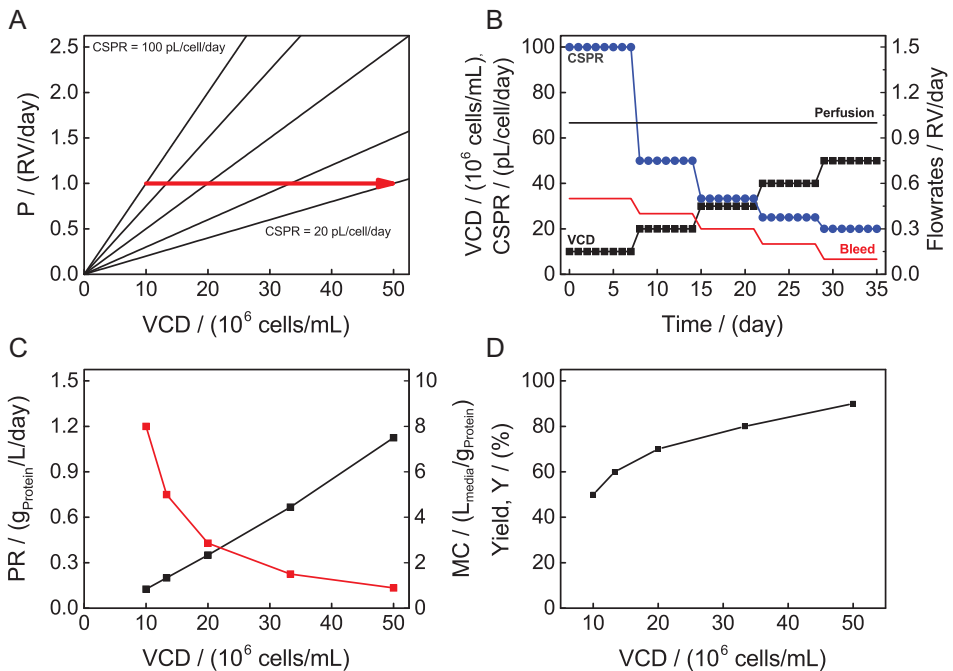


Figure 4.2 Illustration of the first step of the perfusion design: *push-to-high* at constant perfusion rate. For the given example, we consider the operation at a constant perfusion rate of 1.0 RV/day and constant cell-specific productivity of 25 pg/cell/day. Relation between perfusion rate and VCD for various values of the $CSPR$ (A); the arrow shows the decrease of the $CSPR$ (from 100 pL/cell/day down to 20 pL/cell/day), when increasing the VCD at constant P . VCD (■), $CSPR$ (●), perfusion rate (black line) and bleed rate (red line) as a function of time (B). Volumetric productivity (black) and medium consumption (red) as a function of the viable cell density (C). Yield as a function of the viable cell density (D).

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the resulting CSPR ranges between 100 and 20 pL/cell/day. Let us consider the case where the cell-specific productivity is assumed to be constant at 25 pg/cell/day, while the cell-specific growth rate, μ (and, therefore, the bleed rate), decreases with the CSPR. In particular, with illustrative purposes, we assume the cell-specific growth equal to 0.5 day⁻¹ at a CSPR of 100 pL/cell/day and equal to 0.1 day⁻¹ at 20 pL/cell/day (Figure 4.2B). Using Equation (4.7), this results in a titre increasing from 0.25 up to 1.25 g/L for decreasing CSPRs. Finally, using Equations (4.8) and (4.10), we can compute the resulting volumetric productivity, PR , and medium consumption, MC . As shown in Figure 4.2C, we see that the volumetric productivity increases linearly with the viable cell density from 0.13 up to 1.13 g/L/day, while the medium consumption decreases inversely from 8.0 to 0.89 L/g. On the other hand, we see from Equation (4.10) that since the specific cell growth rate decreases, the yield, Y , of the process also improves as the viable cell density increases, as shown in the Figure 4.2D. For this exemplary case, the yield increases from 50 per cent to 90 per cent. It can then be concluded that for the given perfusion rate P , the maximum feasible VCD constitutes the optimal operating condition, which corresponds to the value $CSPR_{min}$.

4.1.3 The Optimal Bioreactor Design Procedure: Step 2

We have seen the process optimisation for a given value of the perfusion rate, P . Now, we want to analyse how the objective functions change when we change P and X_v but keep the CSPR fixed, ideally at its optimum value $CSPR_{min}$. In the case that we express Equations (4.8) and (4.10) as a function of P and CSPR, we obtain the following relations:

$$PR = (P - \mu) \frac{q_p}{CSPR} \quad (4.11)$$

$$MC = \frac{P \times CSPR}{(P - \mu) q_p}. \quad (4.12)$$

If we do the same but now as a function of the viable cell density and the CSPR, we obtain

$$PR = \left(X_V - \frac{\mu}{CSPR} \right) q_p \quad (4.13)$$

$$MC = \frac{X_V \times CSPR}{\left(X_V - \frac{\mu}{CSPR} \right) q_p}. \quad (4.14)$$

If we now consider the operation at constant CSPR (and, therefore, constant q_p and μ), we see from Equations (4.9), (4.11) and (4.12) that as P increases, the productivity and the yield always increase while MC decreases so that we can conclude that the optimum perfusion rate is the maximum value that can be sustained by the set-up under consideration, P_{max} . The same is concluded by considering the viable cell density, as in Equations (4.13) and (4.14). Thus, either the maximum achievable VCD or the maximum sustainable P describe the optimum operating conditions, as illustrated in Figure 4.3 for the illustrative case of a fixed $CSPR = 20$ pL/cell/day. Figure 4.3A shows the relation between VCD and perfusion rate for the selected CSPR value. It is seen that operations at perfusion rates of 1–5.0 RV/day require the set-up to maintain viable cell densities between 50 and 250 $\times 10^6$ cells/mL. (Figure 4.3B). Assuming for this system a constant specific cell growth correspond-

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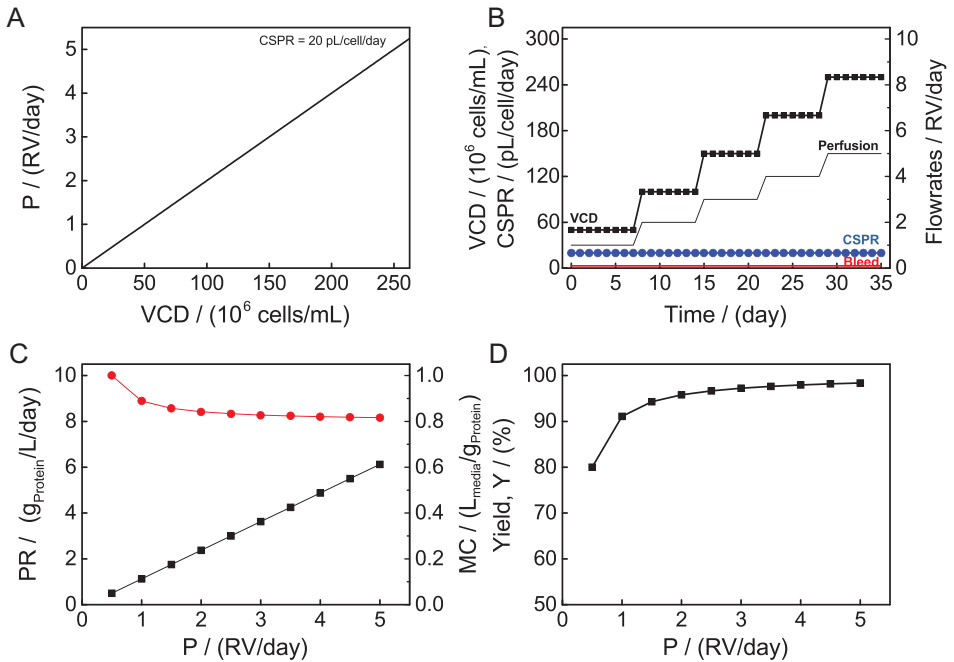


Figure 4.3 Illustration of the second step of the perfusion design procedure at constant CSPR. For the given example, we consider the operation at a constant CSPR of 20 pL/cell/day with a fixed bleed rate of 0.1 RV/day and cell-specific productivity of 25 pg/cell/day. Relation between perfusion rate and VCD for the CSPR value of 20 pL/cell/day (A). VCD (■), CSPR (●), Perfusion rate (black line), and bleed rate (red line) as a function of time (B). Volumetric productivity (black) and medium consumption (red) as a function of the perfusion rate (C). Yield as a function of the perfusion rate (D).

ing to a bleed rate of 0.1 RV/day and a constant cell-specific productivity of 25 pg/cell/day, from Equation (4.7), we compute a constant titre of 1.25 g/L. From this, using Equations (4.9), (4.11) and (4.12), we can calculate the resulting volumetric yield, Y , productivity, PR , and medium consumption, MC , as a function of P . In Figure 4.3C, we can see that the volumetric productivity is increasing linearly with the perfusion rate from 0.5 g/L/day up to 6.1 g/L/day, while the medium consumption decreases inversely from 1.0 L/g to 0.82 L/g. In particular, in the range of low perfusion rates – say, between 0.5 and 3.0 RV/day – increasing the perfusion rate has a strong effect in reducing medium consumption, which becomes less significant at higher P values. On the other hand, it is important to consider the actual feasibility of operating the bioreactor at perfusion rates above 3 RV/day, but especially at viable cell densities above 100×10^6 cells/mL (equal to 2 RV/day at a CSPR of 20 pL/cell/day). In Figure 4.3D, it is seen that the calculated yield increases from 80 per cent to 96 per cent by a step-by-step increase of the perfusion rate from 0.5 RV/day to 2.0 RV/day, and then continues to increase but much slower up to 98 per cent as the perfusion rate increases further. This shows the importance of the optimisation procedure at constant CSPR, as there is a large optimisation potential, but also feasibility constraints that have to be taken into account. The value of the maximum feasible perfusion rate, $P_{\max}$ depends on the expression system, the chosen medium and the reactor.

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From the preceding discussion, we see that the design of a perfusion process requires a sequence of two experimental steps: in the first one, we evaluate the $\text{CSPR}_{\min}$, and in the second, the maximum sustainable perfusion rate, $P_{\max}$ and viable cell density, $X_{V,\max}$ (with ratio equal to $\text{CSPR}_{\min}$), are determined.

4.2 DEVELOPMENT OF A COMMERCIAL PERFUSION CULTURE

The development of a commercial perfusion process for a new cell culture requires a sequence of steps involving experiments ranging from the μL to the L-scale of the laboratory bioreactor and finally to the thousands of litres of the commercial scale. We now introduce the basic concepts of the procedure and next discuss a series of case studies.

The procedure for developing a new commercial perfusion culture is illustrated in Figure 4.4 as constituted of three parts: clone and media screening, perfusion cell culture development and process scale-up. Each one of these parts has its own objectives and requires different technologies, which are discussed separately in the following paragraphs.

Typically, at the beginning of the process development, we have a few hundred clones, coming from the cell line generation and development, that can be combined with several medium compositions for the expression of a given target protein (Part 1). In this first step, the objective is to screen the two or three most promising clones and one or two basic medium compositions in order to proceed with the next step of process development. As objective parameters for the selection, we consider the cell-specific productivity and growth rate, with no reference to product quality or the fact that the culture may be stable or not in perfusion. These two quantities have to be measured for a large number of combinations clone/medium. This requires high-throughput techniques and experiments at the very small scale (μL –mL range).

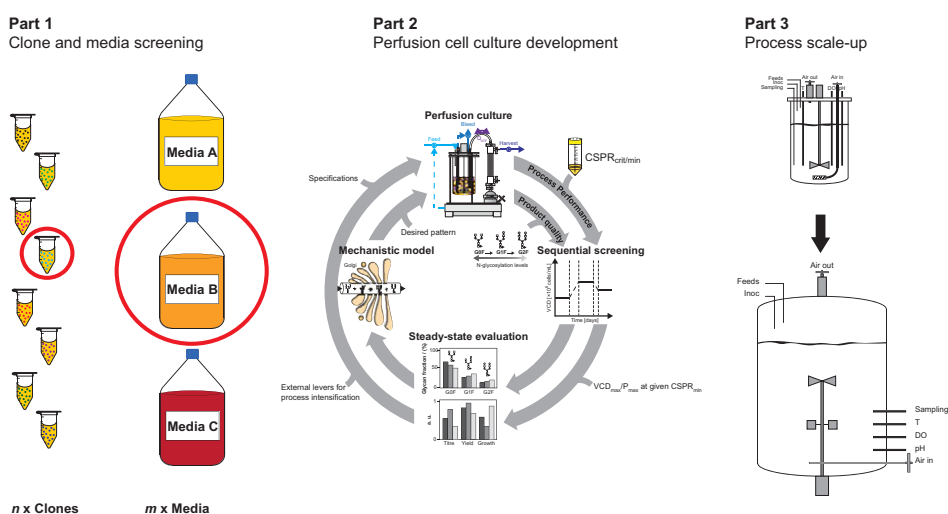


Figure 4.4 Summary of the commercial perfusion culture development procedure: Part 1: Clone and media screening. Part 2: Perfusion cell culture development. Part 3: Process scale-up.

4.3 Part I: Clone and Media Screening for Perfusion Processes

The most common approach is to use scale-down models and particularly high-throughput devices, such as deepwell plate systems. As discussed in Chapter 3, these devices are usually designed for batch or fed-batch operating mode and, therefore, are not best suited for mimicking perfusion operations. Accordingly, in the following, we consider semi-continuous high-throughput units.

Using the few clones and media identified in the first step, we now proceed to identify a set of stable operating conditions for a perfusion bioreactor with optimal performance in terms of maximum productivity and minimum medium consumption. The final result is the determination of the most convenient viable cell density and perfusion rate (Part 2). It has to be mentioned that the product yield does not need to be considered explicitly, since the bleed rate is fixed by the specific cell growth rate of the selected cell line (Equation (4.6)), and therefore, higher productivities imply better yields (Equation (4.9)). Performing all the experiments necessary for this task in the litre scale is too slow and expensive, and therefore, we need again suitable scale-down models that mimic perfusion-like experiments.

In the case that the obtained reactor performance is not satisfactory, it is necessary to go back and to reconsider some of the already selected operating conditions, such as the operating temperature or some chemical additives that may significantly alter the system and then require a new design of the reactor operating conditions. The same situation occurs in the case where the obtained operating conditions do not lead to satisfactory product quality parameters. For example, in the case that the N-linked glycosylation pattern of the product needs to be modified, we have to add some proper feed supplements and then repeat the reactor design procedure.

The objective of the third part is to transfer the perfusion process, developed at the benchtop bioreactor scale, to the larger clinical and commercial scales (Part 3). This means to change the size and shape of the reactor (if needed) and the ancillary equipment, as well as the corresponding operating conditions, in order to reproduce the process performance and product quality attributes obtained at the laboratory scale. At this step, no attempt is made to further improve any of the process or product parameters; the idea is that satisfactory performances have been obtained at the laboratory scale, and now they have to be replicated at larger scales. For this, we have to, for example, make sure that mixing is sufficient, mass transfer does not become limiting and cells do not experience too high shear stress. This is a typical task in chemical engineering, and a whole set of tools has been developed for this in the general area of multiphase reactors, as discussed in Chapter 5.

4.3 PART I: CLONE AND MEDIA SCREENING FOR PERFUSION PROCESSES

As discussed in Chapter 1, cell line development starts with the transfection of the expression vector to the host cell line. This includes the random integration of the foreign genes of interest, which leads to a heterogeneous pool of clones with different growth and protein productivity characteristics. Typically, clones with high and stable recombinant protein production are preferred. However, in a population of transfected cells, only a few clones are high producing, as the protein expression relies on the full integration of the genes for the heavy- and light-chain antibody fragments. As this full integration of the active region supporting high gene expression in the chromosomes of the host cell line is rare (Browne & Al-Rubeai 2007), the selection of proper clones represents a critical and particularly laborious task. In addition, such

high-producer cell clones typically exhibit lower growth rates, as most resources are used for protein expression. Therefore, screening of a very large number of clones is necessary in order to have a good chance to find a few high-producing clones with good cellular growth characteristics (Lai et al. 2013). This step requires high-throughput platforms in order to test and screen large numbers of clones and media at the same time. As mentioned before, the selection criteria are solely based on growth rate and productivity.

Historically, this step is performed in batch mode for both fed-batch and perfusion applications. Typically, cells are sequentially diluted on well plates to screen for high-producer cell clones, thus mimicking a batch-operating mode. However, clones and media that are best for fed-batch may not be optimal for perfusion. For example, a fast growth rate is desired for fed-batch processes in order to accumulate a large number of cells in a short time during inoculation so that these high-producer clones produce as much protein as possible during the production phase. On the other hand, a high growth rate leads to high bleed rates (Equation (4.6)) in order to maintain a constant viable cell density during the steady-state production phase of a perfusion bioreactor and, thus, as shown in Equation (4.9), to low product yields or high product losses. In addition, a very high growth rate triggers high generation numbers and, thus, can lead to cellular instability in perfusion applications operated at very long run durations (Ozturk 2014). Thus, for perfusion cultures, a stable cellular growth rate and high cell-specific productivity are more important than targeting the highest end titre and cell density. In conclusion, we need high-throughput screening devices that can mimic perfusion-like processes and enable long-time screening in order to test the long-term stability and high productivity of the clones of interest.

4.3.1 High-Throughput Systems

Several requirements need to be fulfilled in order to perform cell clone and media screening for a perfusion process. The most critical one is probably to mimic the perfusion operating mode. We can distinguish between two different approaches: non-steady-state and steady-state operation. In non-steady state-perfusion processes, as mentioned in Chapter 2, there is no bleed stream to remove the cells, which, therefore, continuously accumulate in the reactor with respect to their specific growth rate. Accordingly, only medium addition and cell-free harvest are present in this approach. Steady-state perfusion processes, on the other hand, require the cell-removing bleed stream in order to operate at a constant viable cell density, stable in time.

Next we need a scale-down high-throughput system. This should be operated in the nL, μ L scale or low mL scale and should allow parallel operation of several combinations of clones and media at the same time. For this, a high degree of automation is needed in order to reduce manual work. Potential candidate systems for this application are the custom nanofluidic chips using Berkeley Lights OptoElectro Positioning technology, deepwell plates (DWP), shake-tube (ST) bioreactors and the ambr[®] 15 bioreactors, which have been introduced and described in Chapter 3. In order to illustrate the use of such scale-down models for clone screening, we discuss in the following an example using deepwell plates and shake-tube bioreactors and illustrate the validation of the obtained results at the bench scale.

In particular, we consider a semi-continuous procedure, referred to as the VCD_{max} experiment, that allows the operation of perfusion-like processes at constant

4.3 Part I: Clone and Media Screening for Perfusion Processes

medium exchange rate in shake tubes and deepwell plates (Bielser et al. 2019b). The procedure consists of a daily viable cell count, the separation of medium and cells, removal of cell-free supernatant and replacement by fresh medium, without additional cell removal. By repeating this procedure on a daily basis, the cell concentration increases in time until a maximum value – that is, the maximum viable cell density (VCD_{max}) sustainable at the given perfusion rate. This procedure has been shown to work in shake-tube bioreactors in a variety of applications (Gomez et al. 2017, Villiger-Oberbek et al. 2015), and also in deepwell plates (Lin et al. 2017). Bielser et al. (2019b) validated the procedure in shake tubes (15 mL working volume) and 96-DWP (450 μ L working volume) and applied it to screen a set of 12 different CHO cell clones expressing the same IgG1 antibody. The results were compared in terms of VCD_{max} and volumetric productivities.

In the shake tubes, a centrifugation step was applied to separate cells and medium, and the medium in the entire reactor volume was replaced once per day. In the 96-DWP, liquid handling was performed through a robotic platform (Biomek FX, Beckman-Coulter, Fullerton, CA, USA). Here, cells were left to settle for one hour in static conditions in order to separate cells from the medium before half a reactor volume was exchanged. Removing the entire reactor volume was not done due to difficulties in controlling the height of the tip which could lead to additional unwanted cell removal. Cell settling, moreover, results in a less dense cell pellet compared to the centrifugation step applied for the shake-tube system. The results obtained in the ST and the 96-DWP systems were compared, in a first step, in terms of the volumetric productivity at VCD_{max} . The different clones were ranked based on volumetric productivity measured in the two systems. Three groups were identified in each system: high producer (two clones), medium producer (three clones), and low producer (seven clones). The different clones did not share the exact same ranking in the two systems, but at least they were in the same groups, as illustrated in Table 4.1. In a next step, four of these clones were selected to validate the results in benchtop bioreactors at lab-scale (3.5 L working volume).

The clone selection for the benchtop operation was made so as to have sufficient diversity among the different clones: one clone was a higher grower (high VCD_{max}) and gave a high mAb titre in the semi-continuous operation and one clone exhibited low growth (low VCD_{max}) and low mAb titre while the third clone was a high grower but a low producer, and the fourth one was a low grower and a high producer. The results obtained in the two scale-down systems could be validated at lab scale, which, in fact, gave $CSPR_{min}$ and productivity values at steady state similar to those obtained with the VCD_{max} experiments.

Thus summarising, this study demonstrates the potential of the 96-DWP and the shake-tube systems for screening a variety of different clones. In addition, the semi-continuous operation nature of these systems enabled a first prediction of the minimum cell-specific perfusion rate for the considered expression systems. For two of the employed cell clones, the predicted volumetric productivity and cell-specific perfusion rate in the scale-down systems were subsequently validated through long-term benchtop bioreactor runs. The two scale-down systems appear to be very useful for the screening of different media and cell line performances in perfusion mode. The design of the VCD_{max} experiment indeed allows one to make a first estimate of the minimum CSPR but on the other hand, does not allow one to test the cell

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Table 4.1 Ranking of clones 1–12 based on the volumetric productivity values computed from the corresponding mAb concentrations reached at VCD_{max} in ST and 96-DWP. Clones 1 and 5 are highlighted in bold because they were identified as the two top producers in both screening methods. The four clones selected for further investigation in bioreactors were marked with symbol (♦). © Reprinted from *Biotechnology Progress*, Jean-Marc Bielser, Jakub Domaradzki, Jonathan Souquet, Hervé Broly and Massimo Morbidelli, ‘Semi-continuous scale-down models for clone and operating parameter screening in perfusion bioreactors’, 2019, with permission from John Wiley and Sons.

Clone #	Volumetric Productivity in ST (g/L _R /day)	Volumetric Productivity in 96-DWP (g/L _R /day)	Rank in ST	Rank in 96-DWP
#1♦	1.14	0.59	1	2
#5	1.01	0.67	2	1
#6	0.85	0.51	3	4
#7	0.82	0.47	4	5
#2♦	0.67	0.53	5	3
#4♦	0.51	0.28	6	9
#8	0.49	0.18	7	11
#9	0.46	0.45	8	6
#3♦	0.45	0.36	9	7
#10	0.43	0.25	10	10
#11	0.36	0.10	11	12
#12	0.27	0.33	12	8

specific long-term stability. In terms of ease of operation and medium consumption, the DWP system is better than the ST system, but it would still be outperformed by the nanofluidic chip technology presented in Chapter 3.

4.3.2 Considerations on Screening Operation

Clone and medium screening requires, indeed, very small working volumes and needs to take place at the single cell level. We described a variety of scale-down models in Chapter 3, and their selection should be based on the specific objective of the planned experiment and the corresponding stage in process development.

At the very early stage, with hundreds of clones involved, the traditional plate-based workflows and especially the nanofluidic chip device offer the most attractive solutions. As mentioned before, a screening in perfusion mode, which seems to be feasible in the Beacon platform, would be desirable to identify promising candidates for perfusion early. This is definitely done in a second stage, after reducing the number of clones to tens of clones, using the 96-DWP system and the VCD_{max} experiments, which mimic the perfusion operating mode. In this context, it could be of interest to evaluate the 384-DWP systems for such experiments.

The long-term stability testing probably needs to be performed at larger scale, using either the shake-tube system or the ambr[®] 15 system applying sedimentation, or even at larger scale like the ambr[®] 250 perfusion system (Chotteau 2017, Zoro & Tait 2017). The suitability of the different systems for the different stages of cell line and media screening is summarised in Table 4.2.

4.4 Part 2: Perfusion Cell Culture Development

Table 4.2 Comparison of various high-throughput systems with respect to their suitability for the different stages of the cell line and media screening processes. A plus sign indicates the system is promising, a minus sign indicates the system is not suited for the specific task.

Application	Beacon platform	96- / 384-DWP	Shake tubes (10–20 mL)	ambr15	250 mL bioreactors
Cell line and media screening	+	+	–	–	–
Suitability for perfusion	+	+	+	+	+
Long-term stability	+	–	+	+	+

4.4 PART 2: PERFUSION CELL CULTURE DEVELOPMENT

In this step, we identify, for the few selected clone/medium combinations, the best operating conditions for a stable perfusion process at the litre scale targeting maximum productivity ($g_{\text{protein}}/\text{L}/\text{day}$) and minimum medium consumption ($L_{\text{medium}}/g_{\text{protein}}$). Already at the litre scale, perfusion processes require large amounts of medium. Operating a perfusion process in a 2 L reactor at a perfusion rate of 1 RV/day for 30 days requires 60 L of perfusion medium. If one considers the number of experiments that have to be performed in order to determine optimum operating conditions, it appears that the whole procedure would require a long time and large medium volumes. Therefore, we need to use scale-down models that allow operation at the μL or mL scale.

The general procedure for the optimal design of a perfusion bioreactor, in terms of perfusion rate and viable cell density, has been described in Section 4.2 based on the two steps illustrated in Figure 4.5: (1) determination of the minimum cell-specific perfusion rate and (2) the determination of the best P (or VCD) for the given CSPR_{min} . The CSPR_{min} describes the minimum possible value of the CSPR to have a sustainable perfusion culture. Therefore, for a given expression system (medium and clone), at CSPR_{min} the cellular growth cannot be further decreased and cell-specific productivity cannot be further increased without changing media or clones.

The reactor performance obtained at this point may still not be satisfactory and, therefore, requires some further optimisation steps. We consider, as examples, the two further steps illustrated in Figure 4.5. The first one refers to the case, where the specific cellular growth is still high at the identified CSPR_{min} and therefore leads to excessive product losses in the bleed. In order to improve the process yield, it is then possible to introduce changes in the operating conditions to inhibit cell growth (Lin et al. 2017, Wang et al. 2018, Wolf et al. 2019a).

The fourth step refers to product quality. This has to satisfy certain predefined specifications and in some cases, such as for biosimilars, it has to be sufficiently similar to a desired quality pattern. When this is not the case, we need to take action, which is mostly done through the introduction of proper nutrient additives. In Section 4.5.3 and in particular in Chapter 6, we discuss the combination of mechanistic modelling and the experimental investigations to modulate certain product quality attributes through proper continuous feed additions (Karst et al. 2017b; Villiger et al. 2016a, 2016b).

4 Design and Optimisation of Mammalian Cell Perfusion Cultures

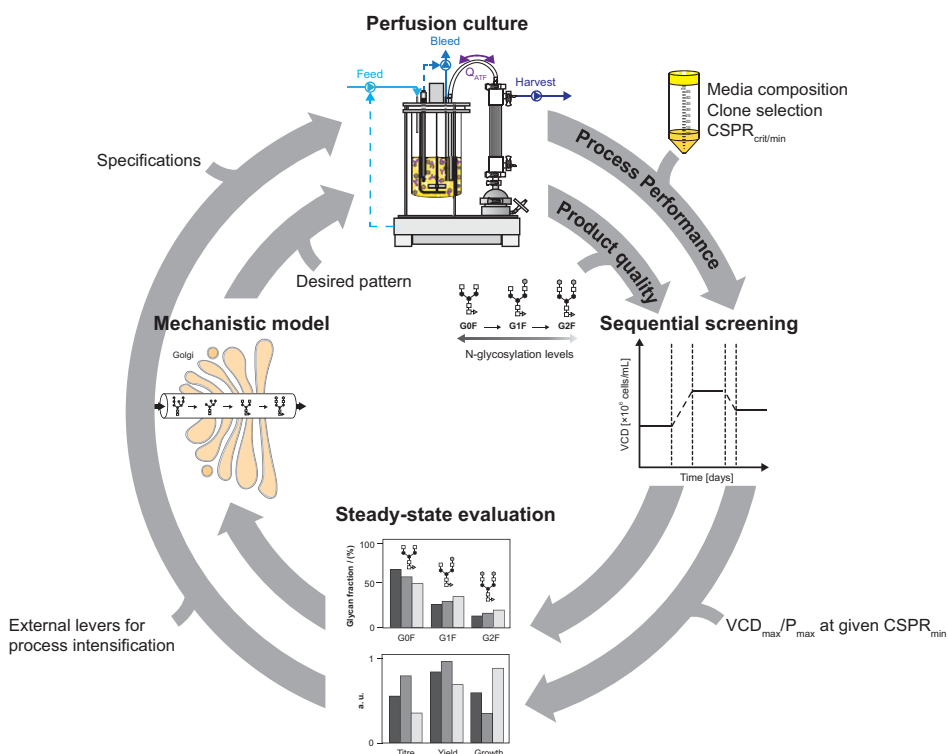


Figure 4.5 Schematic of the procedure for the optimal bioreactor design. In the first step, the minimum CSPR is identified, and in the second, the maximum volumetric productivity at the CSPR_{min} is defined. In the third step, the process yield is optimised through cell-growth inhibition, and in the last, product quality is modulated to match a desired pattern. The procedure is iterated until converging to satisfying overall reactor performance.

At the end of both the third and the fourth steps, the procedure for the design of the perfusion rate and the viable cell density has to be repeated, thus leading to the iterative procedure illustrated in Figure 2.5, which ends when all aspects of the perfusion culture are considered satisfactory. The scope of the following sub-sections is to provide examples of application of this procedure, involving in all cases systems of commercial interest.

4.4.1 Definition of Minimum CSPR

Being the first step of the procedure, this is certainly the most challenging one, since we still have very little knowledge about the expression system. Minimising the CSPR can be done by applying the push-to-low/push-to-high approach shown in Figure 4.1. This is done in one experiment where, starting from a steady-state at a relatively high CSPR value, the bioreactor is forced to go through a sequence of steady states characterised by decreasing CSPR values. As described in Figure 4.1, this can be done by either step wise decreasing P , while keeping VCD constant, or decreasing stepwise VCD with P unchanged. These experiments at the litre scale require very long times and the consumption of relevant amounts of media, thus calling for scale-down models in the mL or μ L scale.

4.4 Part 2: Perfusion Cell Culture Development

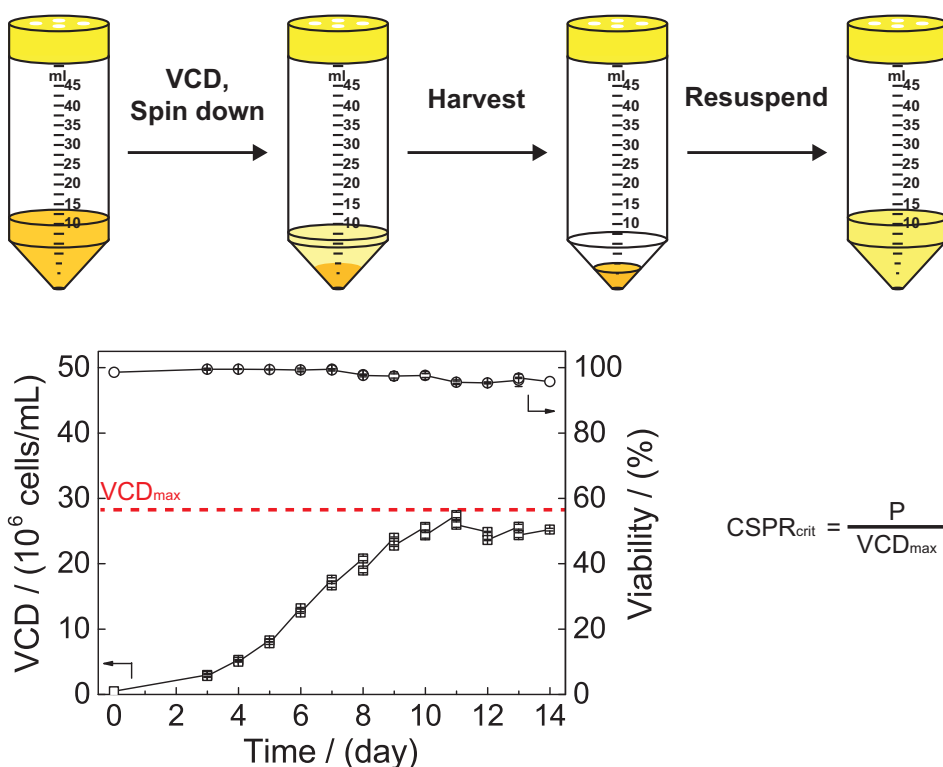


Figure 4.6 Schematic representation of the VCD_{max} experiment. After inoculation at low VCD (e.g. 0.5×10^6 cells/mL), STs are operated in a semi-continuous way. Every 24 h, cells are counted, spun down, the cell-free supernatant harvested, replaced by fresh medium and the cells resuspended. This operation results in the maximum viable cell density achievable at the given medium exchange rate, and allows determining the critical value CSPP_{crit}. ©Reprinted from *Biotechnology and Bioengineering*, Moritz K. F. Wolf, Andrea Müller, Jonathan Souquet, Hervé Broly and Massimo Morbidelli, 'Process design and development of a mammalian cell perfusion culture in shake-tube and benchtop bioreactors', 2019, with permission from John Wiley and Sons

VCD_{max} Experiments: Evaluation of CSPP_{crit}. The VCD_{max} experiments described in Section 4.4.1 and used for clone/media screening in DWP (μL) and ST (mL) units provide the first step for the procedure to define CSPP_{min}. The procedure consists of a daily viable cell count, the separation of medium and cells, removal of cell-free supernatant and medium replacement by fresh medium, before either shake tubes or deepwell plates are transferred back to the incubator without having removed any cell, as illustrated in Figure 4.6. By repeating this procedure on a daily basis, the maximum viable cell density (VCD_{max}) achievable at the given perfusion rate can be evaluated. This corresponds to the viable cell density value where the provided amount of nutrients per cell per day (CSPP) becomes critical and cellular density stops increasing or even starts decreasing, as shown in Figure 4.6. Note that this type of experiment does not include cell removal (bleeding) and, therefore, cannot assess the long-term stability of the perfusion run. Accordingly, we refer to this as the critical CSPP, CSPP_{crit}, which should not be confused with the CSPP_{min}.

In the case of Bielser et al. (2019b), the evaluated CSPP_{crit} in DWP and ST was found to lead to long-term stable perfusion runs in the bioreactor, and

therefore, it was not distinguished from the $\text{CSPR}_{\min}$. However, in the case of Wolf et al. (2019b), the viable cell density values measured in these experiments – i.e., $\text{VCD}_{\max} = 27.5 \times 10^6$ cells/mL – could not be maintained in the long-term bioreactor operation, for which a lower value – i.e., $\text{VCD} = 20 \times 10^6$ cells/mL – had to be used. Such a different behaviour can indeed be attributed to different cell lines used in the two experiments. In all cases, it is recommended not to rely only on experiments at this small scale for the definition of the $\text{CSPR}_{\min}$ but to validate it at a larger scale with long-term stability experiments at constant viable cell density, thus equipment allowing for cell removal.

It has to be noted that, if the identified $\text{VCD}_{\max}$ corresponds to the complete depletion of one or more nutrients in the used medium, a medium with higher depth can be prepared, and the procedure can be repeated, as discussed for the push-to-low and push-to-high steady-state experiments in Konstantinov et al. (2006). It should be considered that this may lead to higher osmolality values in the reactor, which needs to be accounted for.

Steady-State Experiments in the mL Scale. Once the $\text{VCD}_{\max}$ and $\text{CSPR}_{\text{crit}}$ values have been obtained from the perfusion scale-down models discussed earlier, we need to proceed to identify the $\text{CSPR}_{\min}$ – that is, to identify the minimum sustainable CSPR at which a stable culture can be operated for longer times. As mentioned before, such a $\text{CSPR}_{\min}$ value can be either equal to or larger than the $\text{CSPR}_{\text{crit}}$, which therefore provides an excellent starting point. This can be done either directly in the bioreactor at the litre scale or, even better, in a similar scale-down model.

Wolf et al. (2018) developed a shake-tube-based scale-down model in order to investigate steady-state operations in the mL scale, referred to in the following as VCD_{ss} . The developed procedure is similar to the one for the $\text{VCD}_{\max}$ experiment but with two significant differences. Firstly, the shake tubes are inoculated at very high cell densities close to the targeted viable cell density. Secondly, in order to operate at a constant viable cell density, a bleed step is introduced. This comes in addition to the cell removal connected with the culture sampling for the cell count, and it is specifically designed to keep the VCD constant around a predefined target value ($X_{V,\text{target}}$), as illustrated in Figure 4.7. The bleed and harvest rates are mimicked by removing cell-containing and cell-free volumes, respectively, from the ST, which is calculated on a daily basis after the cell count. The first one has to be computed so that the VCD value remains close to $X_{V,\text{target}}$ through all the cycles. In particular, we have to remove a certain cell-containing volume in order to decrease the VCD (after fresh medium addition) to a value, $X_{V,SP}$, such that the average between $X_{V,SP}$, and the measured VCD, $X_{V,\text{meas}}$, is equal to $X_{V,\text{target}}$. Accordingly, based on the measured VCD, $X_{V,\text{meas}}$, the bleed volume can be calculated as follows:

$$V_{\text{Bleed}} = \frac{(X_{V,\text{meas}} - X_{V,SP}) V_{\text{Tot}}}{X_{V,\text{meas}}}, \quad (4.15)$$

where V_{Tot} represents the total working volume in the ST and V_{Bleed} is the total cell-containing volume to be removed from the ST including the volume of the sample taken for the cell count. The harvest volume is consequently calculated as follows:

$$V_{\text{Harvest}} = V_{\text{Exchange}} - V_{\text{Bleed}}, \quad (4.16)$$

where V_{Exchange} represents the targeted volume that should be replaced with fresh medium to simulate a given perfusion rate. For example, in the case of perfusion rates

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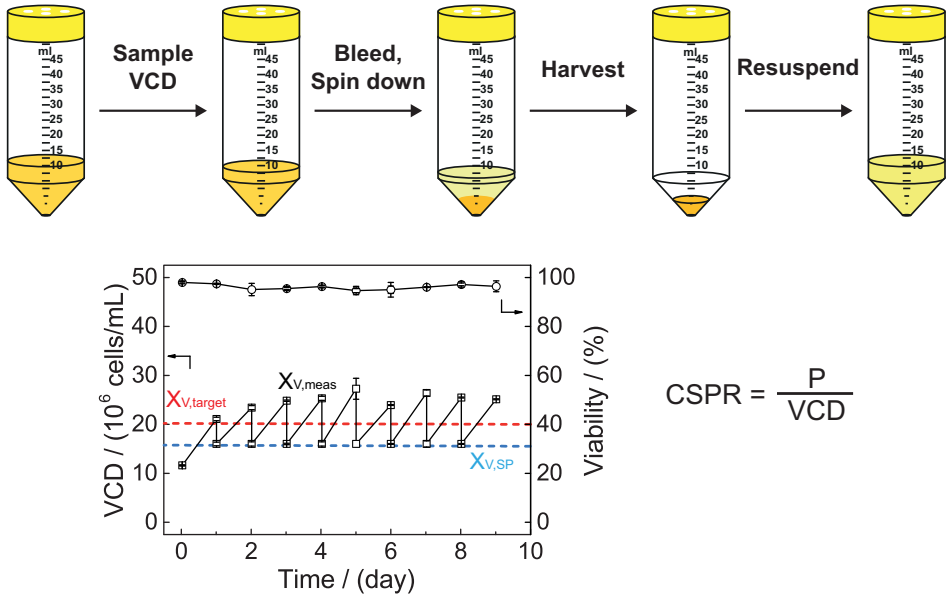


Figure 4.7 Illustration of the VCD_{ss} experiment mimicking a perfusion culture at predefined VCD, $X_{V,target}$ and constant perfusion rate, P . Cells are inoculated close to the VCD bleeding set point ($X_{V,SP}$), then STs are operated in a semi continuous way. Every 24 hours, cells are withdrawn and counted. Some cells are removed based on the measurement and the remaining cells are spun down. The remaining cell-free supernatant is harvested, and the cell pellet is resuspended in fresh medium. The plot shows the VCD value oscillating above and below the predefined value, $X_{V,target}$, at the given medium exchange in the case of a stable operation – i.e., CSPR larger or equal CSPR_{min}. © Reprinted from *Biotechnology and Bioengineering*, Moritz K. F. Wolf, Andrea Müller, Jonathan Souquet, Hervé Broly and Massimo Morbidelli, 'Process design and development of a mammalian cell perfusion culture in shake-tube and benchtop bioreactors', 2019, with permission from John Wiley and Sons

of one reactor volume per day, $V_{Exchange}$ is equal to one reactor volume, that is, to V_{Tot} . The flowrates in the corresponding continuous perfusion are then calculated as follows:

$$B = \frac{V_{Bleed}}{V_{Tot}} \frac{1}{\Delta t} \quad (4.17)$$

$$H = \frac{V_{Harvest}}{V_{Tot}} \frac{1}{\Delta t} \quad (4.18)$$

This scale-down model has been compared with a benchtop bioreactor operated at similar VCDs and perfusion rates (Figure 4.8A). It is shown that, not only are the bleed and harvest rates (Figure 4.8B) in the two systems very similar, but the entire culture develops along similar lines. This includes metabolite consumption or production rates, such as glucose and ammonia, cell-specific growth rates, titre and even the charge variant pattern of the target product (Figure 4.8C, D and F). On the other end, it is worth mentioning that the ST model differed in the lactate production and the product glycosylation pattern observed in the bioreactor (Figure 4.8C and E). This may be because of both the decreased removal of CO₂ and the lower dissolved oxygen levels due to the smaller gas–liquid mass transfer rate observed

4 Design and Optimisation of Mammalian Cell Perfusion Cultures

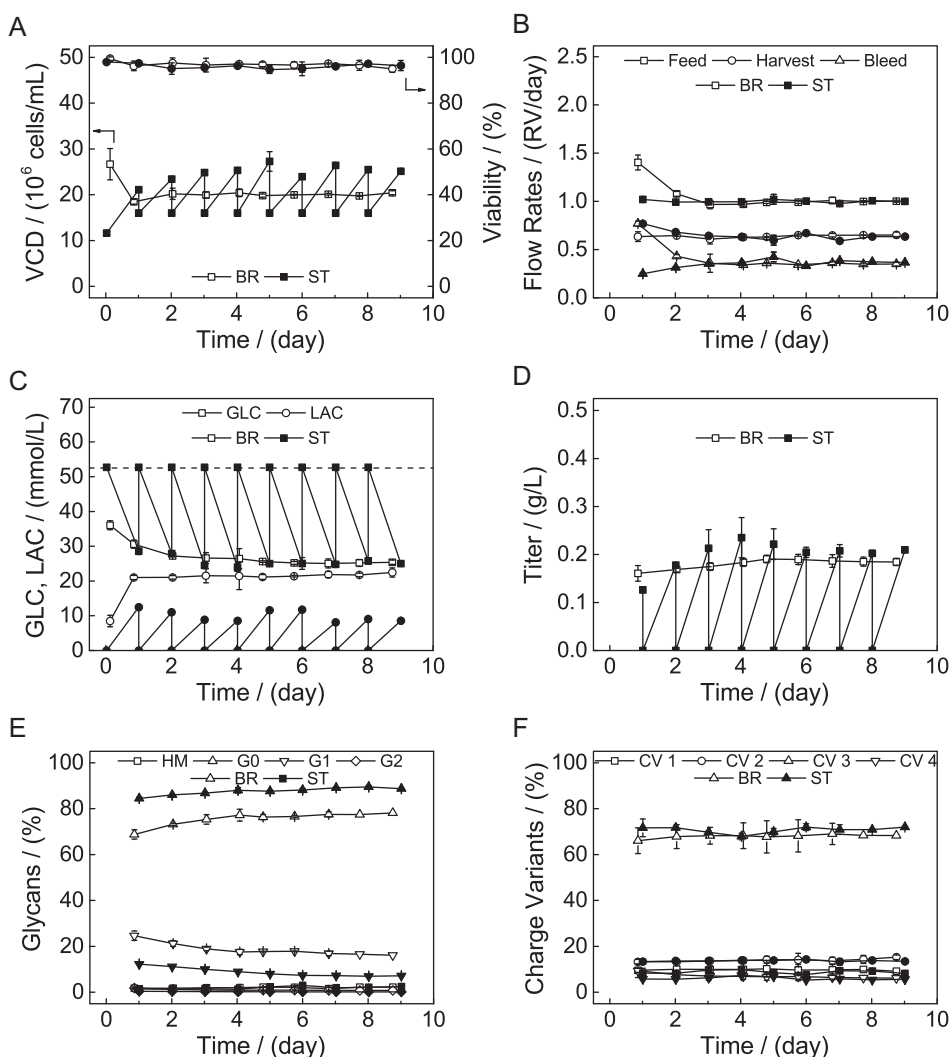


Figure 4.8 Comparison of a steady-state perfusion cell culture performed in benchtop BR and semi-continuous shake-tube (ST) bioreactors at a VCD set point of 20×10^6 cells/mL. VCD and viability (A), process exchange flowrates (B), glucose and lactate concentrations (C), product titre (D), N-linked glycoforms (E) and charge variants (F). © Reprinted from *Biotechnology and Bioengineering*, Moritz K. F. Wolf, Veronika Lorenz, Daniel J. Karst, Jonathan Souquet, Hervé Broly and Massimo Morbidelli, 'Development of a shake-tube based scale-down model for perfusion cultures', 2018, with permission from John Wiley and Sons

in the ST. Here this transfer occurs through the filter in the cap of the shake tube and is therefore less efficient than the gas spargers used in benchtop bioreactors. This results in more acidic conditions, to which the cells counteract by producing less lactate to maintain the pH at acceptable levels. Concerning the glycan distribution, the less controlled oxygen environment and the difference in oxygenation are expected to result in lower dissolved oxygen concentration in the ST compared to the benchtop system. Lower DO levels are reported to affect the mAb glycosylation,

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such that the level of galactosylated glycans decreases (Kunkel et al. 2000; 1998). Besides this aspect, the developed procedure enabled the mimicking of steady-state perfusion processes in bioreactor at the benchtop scale and can therefore be used for designing their operating conditions, at least in terms of perfusion rate and viable cell density.

In conclusion, it is convenient to combine the VCD_{max} experiment described earlier with the VCD_{ss} procedure, in order to first estimate a suitable VCD_{max} and the corresponding $CSPR_{crit}$, which is then taken as a starting point to determine the $CSPR_{min}$. This approach is very advantageous in practice since the simple VCD_{max} experiment easily provides a first impression about the behaviour of the selected expression system (clones and media) at a given perfusion rate and is, therefore particularly, suitable for screening applications. This experiment results in a first guess of the $CSPR_{min}$, referred to as the $CSPR_{crit}$, which needs to be further elaborated to obtain the $CSPR_{min}$ with respect to long-term stability using, for example, a benchtop bioreactor. On the other hand, the VCD_{ss} experiments can be used for the same objective of defining the $CSPR_{min}$, also considering that they provide reliable estimates of various process parameters in the bioreactor, such as the VCD and the harvest and bleed flowrates at a given perfusion rate, as well as cell-specific metabolism and a first prediction of product quality patterns. As these experiments can be performed in DWP (VCD_{max}) and ST (VCD_{max} and VCD_{ss}), they need only very small amounts of media and can be performed in parallel, which therefore results in significant cost and time savings. It can be expected that even better scale-down models could be obtained by realising more automated and better controlled systems that allow continuous media exchange at a small scale, in the direction of the work done by Chotteau (2017) and Zoro & Tait (2017) at the 250 mL scale. Alternatively, it is also possible to perform the push-to-low and push-to-high experiments in lab-scale bioreactors (1–10 L), as illustrated by Konstantinov et al. (2006) and Wolf et al. (2019c). In such a case, a sequential bioreactor run is performed, as illustrated in Figure 4.2B. However, this operation requires much higher media volumes compared to scale-down models.

4.4.2 Optimisation of Volumetric Productivity at $CSPR_{min}$

After determining the $CSPR_{min}$, the process can be further improved in terms of volumetric productivity and medium consumption, as discussed in Section 4.1.2. This can be done in either ST or benchtop bioreactors with increasing P or VCD while keeping constant $CSPR = CSPR_{min}$. By operating at constant $CSPR$, the cellular activity in terms of q_p and μ remains constant (Ozturk 2014), and then, as shown by Equations (4.9), (4.11) and (4.12), the volumetric productivity and the yield increase and the medium consumption decreases, as the perfusion rate increases. In addition, the average product residence time decreases, leading to a general improvement of the product-related quality attributes. Thus, overall, as the perfusion rate increases, the bioreactor performance improves in all aspects. This, however, cannot be continued indefinitely since, at some point, some limitation is encountered because of mixing intensity, oxygen supply, carbon dioxide removal or something else.

This has been illustrated by Wolf et al. (2018) in a series of VCD_{ss} experiments at 20 and 40×10^6 cells/mL with perfusion rate of 1 and 2 RV/day, respectively, so as to have the same value of $CSPR$ of 50 pL/cell/day. To match a perfusion rate of

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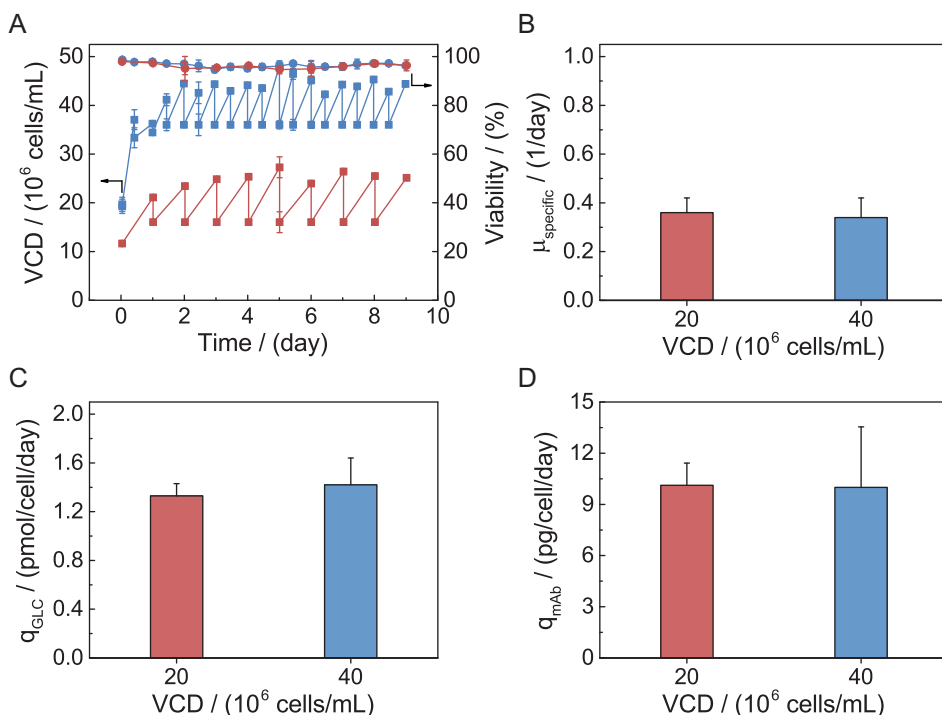


Figure 4.9 Viable cell density and viability (A) as a function of time, cell specific growth rate (B), glucose consumption rate (C) and cell-specific mAb production rate (D) as a function of the viable cell density for two experiments at the same CSPP of 50 pL/cell/day in a ST system. © Reprinted from *Biotechnology and Bioengineering*, Moritz K. F. Wolf, Veronika Lorenz, Daniel J. Karst, Jonathan Souquet, Hervé Broly and Massimo Morbidelli, ‘Development of a shake-tube based scale down model for perfusion cultures’, 2018, with permission from John Wiley and Sons

2 RV/day in the ST system, two full media exchanges per day – that is, every 12 instead of 24 h – was performed. As expected, similar cell-specific growth rates, glucose consumption rates and specific antibody production rates were found, as shown in Figure 4.9. On the other hand, the process performance improved in terms of productivity, medium consumption and yield, as shown in Figure 4.10.

As the ST system is quite limited, especially in terms of oxygen supply and carbon dioxide removal, it is generally not possible to operate steady-state experiments at viable cell densities above 50×10^6 cells/mL (Villiger-Oberbek et al. 2015, Wolf et al. 2018). Thus, we can follow an alternative approach and perform this step of the design procedure in a benchtop-reactor system.

The preceding results were confirmed using a benchtop bioreactor, operated with the same expression system and again at the same CSPP = 50 pL/cell/day, as shown in Figure 4.11 (Wolf et al. 2019c). A sequence of three steady-state experiments were conducted at VCD values of 20, 40 and 30×10^6 cells/mL (Figure 4.11A), with perfusion rates of 1, 2 and 1.5 RV/day, respectively (Figure 4.10B), in order to maintain the constant CSPP value (Figure 4.11C). As a result, the cells exhibited constant growth rate resulting in a constant bleed rate (Figure 4.11B) and constant metabolite environment (Figure 4.11D and E). The same behaviour is exhibited by the glycoform distribution shown in Figure 4.11F as a function of time, which remains substantially

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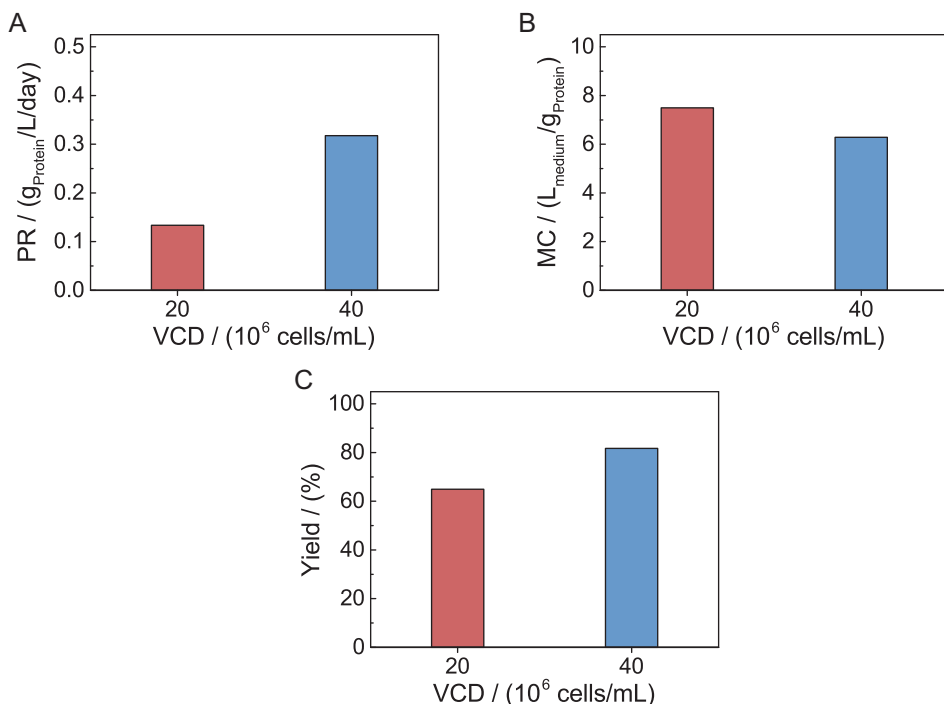


Figure 4.10 Volumetric productivity (A), medium consumption (B) and harvest yield (C) as a function of the viable cell density for two experiments at the same CSPR of 50 pL/cell/day in a ST system.

constant across the three steady-states. Based on these results and Equation (4.7), one should also expect a constant antibody concentration, which, however, as seen in Figure 4.12, was not the case.

This anomaly is due to an intrinsic instability of the used cell clone, which has been consistently observed in various experiments with the same cell line (Karst et al. 2016, Steinebach et al. 2017, Wolf et al. 2019c), which leads to the decrease of the cell-specific productivity, q_p , in time shown in Figure 4.12.

Unfortunately, this masked the effect of perfusion rate on productivity and media consumption, as apparent in Equations (4.11) and (4.12), respectively. However, the specific growth rate was apparently not affected by this cell intrinsic instability and therefore, according to Equation (4.9), the effect of P on the process yield, Y could still be observed correctly. In fact, as shown in Figure 4.11B, the bleed rate remained pretty constant during the experiment as the perfusion rate increased from 1.0 to 1.5 and then to 2.0 RV/day. This resulted in a stepwise increase of the yield from 62 per cent to 85 per cent, as shown in Figure 4.13.

The examples discussed so far illustrated the procedure for the optimal bioreactor design developed in the two steps in Section 4.2. We now proceed with the further development of the culture process in the case where the performance obtained at this point was not satisfactory for some reason, such as low yield or large media consumption or others. A good example is the case seen earlier, where the selected expression system exhibits undesired long-term instability or ageing. Some methods to further improve the process performance are discussed in the next section.

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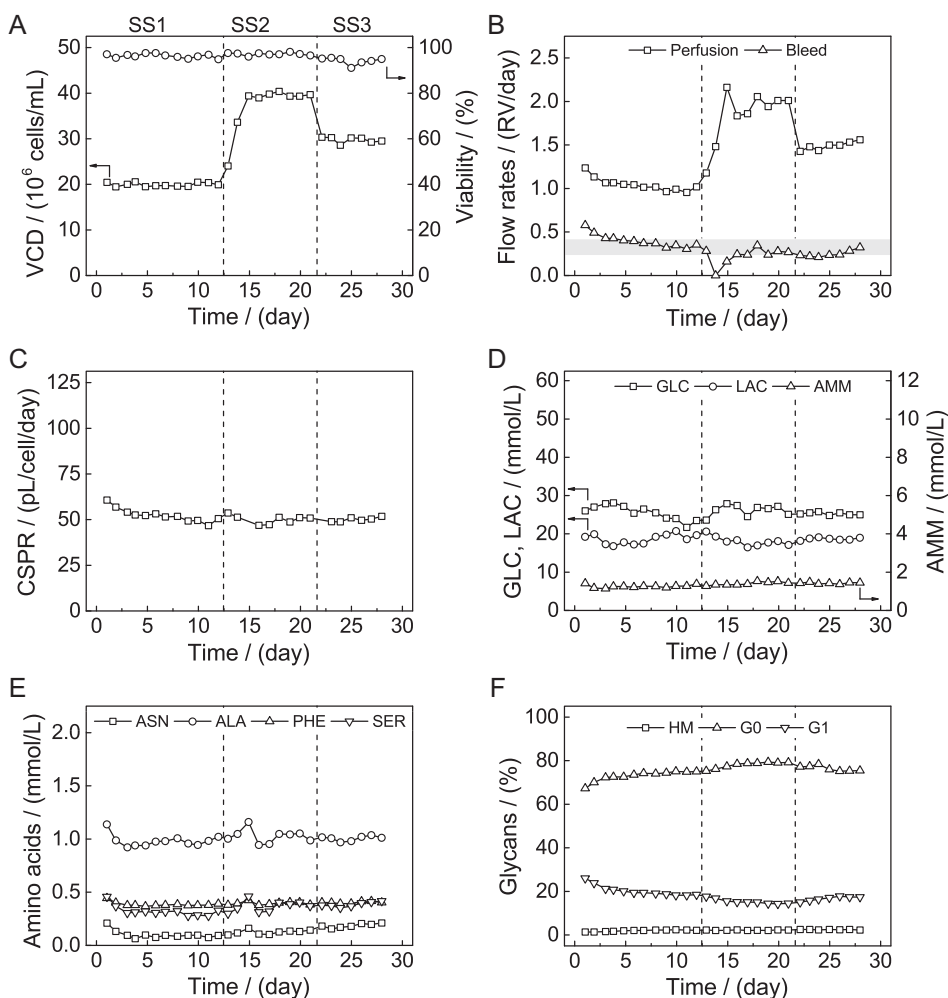


Figure 4.11 VCD and viability (A), perfusion and bleed rate (B), CSPR (C), GLC, LAC, AMM concentrations (D), amino acid concentrations (E) and glycan distribution (F) as a function of time in a sequence of three steady-state experiments at constant CSPR = 50 pL/cell/day. © Reprinted from *Biochemical Engineering Journal*, 151, Moritz K. F. Wolf, Anna Pechlaner, Veronika Lorenz, Daniel J. Karst, Jonathan Souquet, Hervé Broly and Massimo Morbidelli, 'A two-step procedure for the design of perfusion bioreactors', 2019, with permission from Elsevier

4.4.3 Further Actions for Improving Process Performance

As mentioned before, clones best suited for fed-batch culture should exhibit high cell-specific growth rates and, in particular during the stationary phase, high cell-specific productivities. On the other hand, considering a steady-state perfusion process at CSPR_{min}, cells should exhibit low cell-specific growth, thus resulting in high process yields and high and stable cell-specific productivities. As perfusion processes last longer than fed-batch processes, more consistent and stable cellular activities are required. Accordingly, in several studies, researchers have investigated the possibility to improve the performance of a fed-batch-platform expression system when

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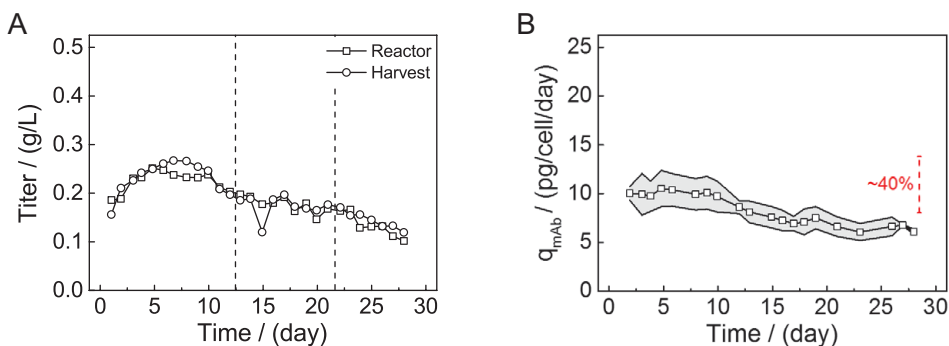
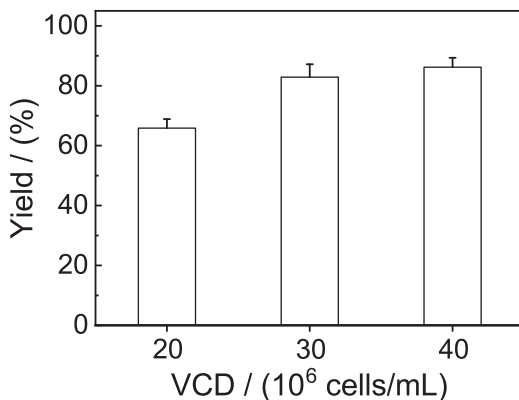


Figure 4.12 Titre in reactor and harvest (A) as a function of time in the sequence of three steady-state experiments at constant CSPR = 50 pL/cell/day in Figure 4.10, and historically observed decline in the cell specific productivity observed in these and other experiments. © Reprinted from *Biochemical Engineering Journal*, 151, Moritz K. F. Wolf, Anna Pechlaner, Veronika Lorenz, Daniel J. Karst, Jonathan Souquet, Hervé Broly and Massimo Morbidelli, 'A two-step procedure for the design of perfusion bioreactors', 2019, with permission from Elsevier

Figure 4.13 Yield as a function of the viable cell density in a sequence of three steady-state experiments at constant CSPR = 50 pL/cell/day.



used in a perfusion bioreactor. These include the implementation of chemical and environmental growth inhibition and rebalancing of the medium composition based on the investigated energy metabolism of the cells (Du et al. 2015, Lin et al. 2017, Wang et al. 2018, Wolf et al. 2019a). These studies target especially active growth inhibition as soon as the production phase is reached, in order to reduce cellular growth and trigger target protein production.

In this frame, Wolf et al. (2019a) considered the application of induced cell-cycle inhibition for cellular growth control. The cell cycle comprises four different phases: G0/G1, S, G2 and M (Pardee 1989, Watanabe et al. 2002). The progression between the different phases is controlled by specific complexes of cyclins and cyclin-dependent kinases. Arresting the cell cycle in the G0/G1 phase would be particularly convenient since, in addition to inhibiting cell doubling, this would also maximise the protein production metabolism of the cell (Barberis et al. 2007, Sherr & Roberts 1999). For this, Wolf et al. (2019a) tested valeric acid, a short-chain fatty acid reported to be a promising G0/G1 inhibitor and productivity enhancer, as a chemical

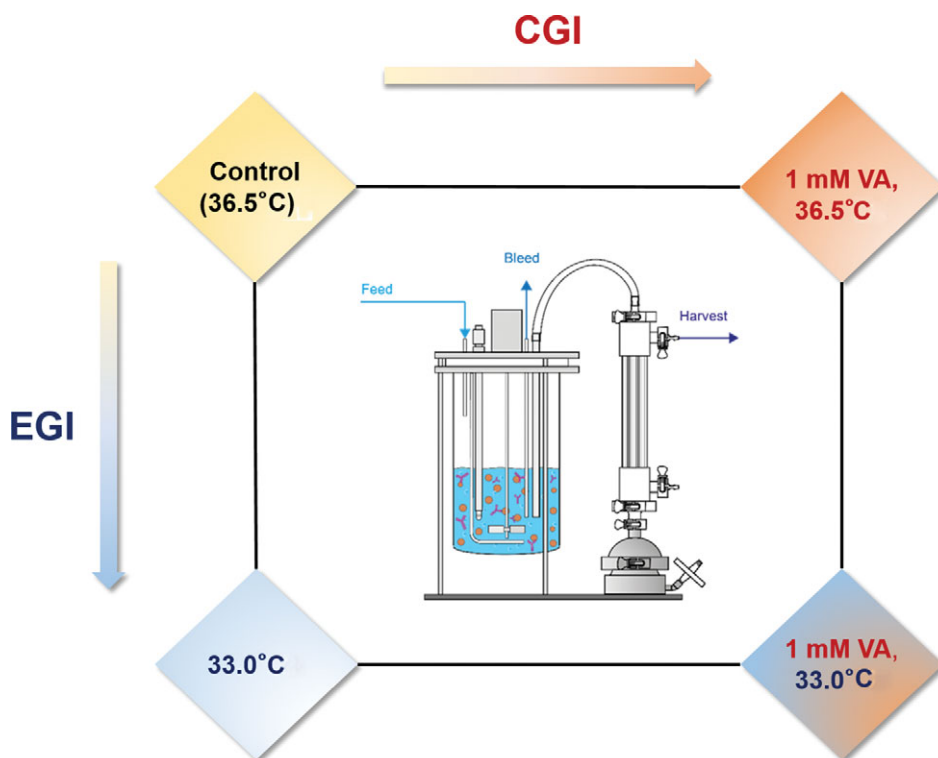


Figure 4.14 Experimental design for the investigation of the effect of environmental (EGI) growth inhibition and chemical growth inhibition (CGI).

growth inhibitor (CGI). In addition, mild hypothermia was also considered as an environmental growth inhibitor (EGI) because of its potential to reduce cell growth (Chuppa et al. 1997, Coronel et al. 2016, Kaufmann et al. 1999, Park et al. 2015). Four experiments were designed in order to investigate the effect of low temperature and valeric acid during the production phase of a perfusion bioreactor.

In particular, as shown in Figure 4.14, these included, after high cell density seeding from an N-1 perfusion bioreactor, the control culture at 36.5°C without valeric acid, a culture at the same temperature but with the addition of 1 mM valeric acid to the perfusion medium (CGI), a culture at 33.0°C (CGI) and a culture with both CGI and EGI – that is, 33.0°C – and with the same amount of valeric acid. In all cases, the perfusion runs were operated at VCD of 30×10^6 cells/mL and with a harvest rate of 1 RV/day, so the overall perfusion rate depended on the extent of growth inhibition and the resulting bleed rate. In all cases two-stage perfusion cultures were operated. Stage 1 represents the N-1 perfusion bioreactor, where cells exhibit high cell specific growth rates to rapidly achieve high cell numbers, while stage 2 represents the production phase. The separation of the two stages allows the time to steady state within the production bioreactor to be shortened.

While the combination of environmental and chemical growth inhibition did not allow a stable reactor operation, stable cell cultures were obtained at control, environmental growth inhibition and chemical growth inhibition conditions, as

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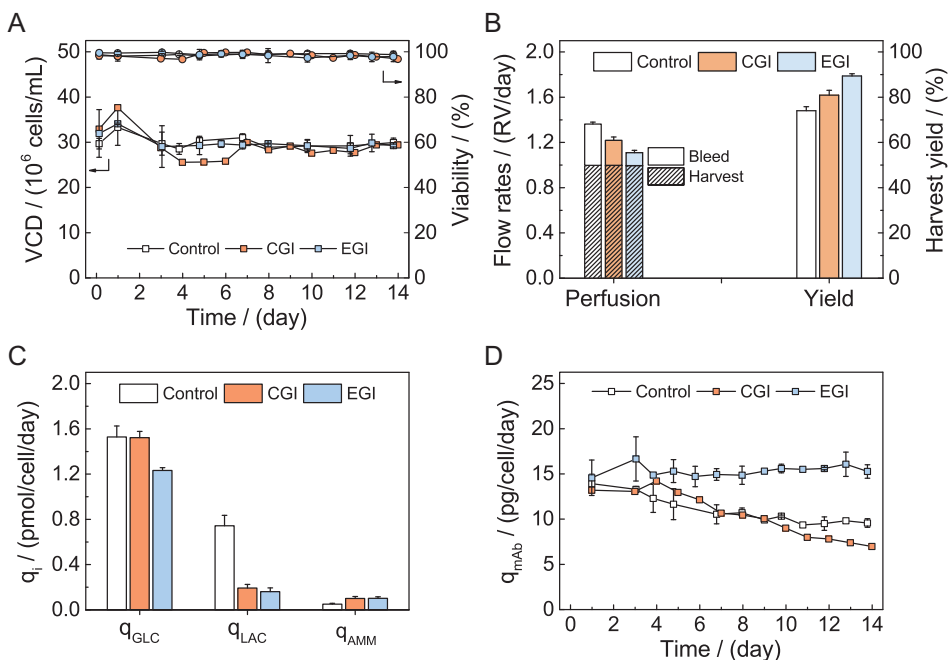


Figure 4.15 Results of the growth inhibition study. (A) VCD and viability as a function of time. (B) Flowrates and process yield for the different process conditions. (C) Cell specific rates for the different process conditions (D) Cell-specific productivity as a function of time. © Reprinted from *Biotechnology Journal*, Moritz K. F. Wolf, Aurélie Closet, Monika Bzowska, Jean-Marc Bielser, Jonathan Souquet, Hervé Broly and Massimo Morbidelli, 'Improved performance in mammalian cell perfusion cultures by growth inhibition', 2019a, with permission from John Wiley and Sons

illustrated in Figure 4.15A. The application of growth inhibition resulted in both cases in reduced bleed rates – in particular, in the case of environmental growth inhibition (Figure 4.15B). This led to a bleed rate of 0.1 RV/day, compared to 0.4 RV/day at control conditions, resulting in a process yield improvement from 74 per cent to almost 90 per cent. The different strategies affected the cellular activity. While cells exhibited similar glucose consumption under chemical growth inhibition and control conditions, under environmental inhibition conditions, cells showed a lower glucose consumption rate (Figure 4.15C). Nevertheless, under environmental and chemical growth inhibition conditions, cells showed a similar and much lower lactate production rate and also a similar but slightly higher ammonia production rate compared to the control conditions (Figure 4.15C). Considering the specific antibody productivity, cells showed the same instability in the specific protein productivity observed previously under control conditions (Karst et al. 2017a). This same trend was observed, when cells were cultivated in the presence of valeric acid (Figure 4.15D). Instead, under environmental growth inhibition, this cell-specific instability did not occur, and cell-specific mAb productivity was slightly enhanced, but more importantly, it became stable.

By investigating the cell cycle arrest, it was found (Figure 4.16) that both inhibition strategies led to cell-cycle arrest in the G0/G1 phase compared to the control condition. While environmental growth inhibition led to a significant and constant

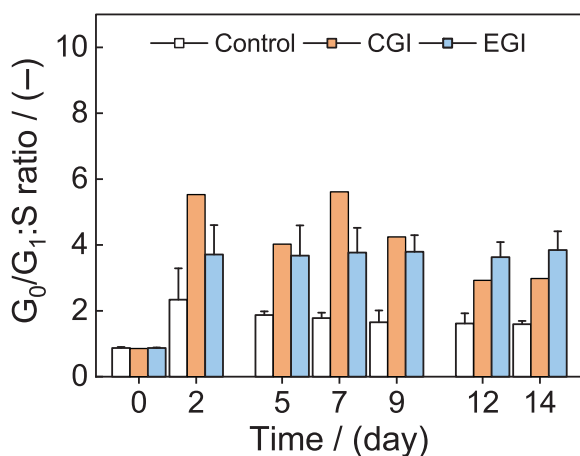


Figure 4.16 The G₀/G₁:S ratio represents the extent of cell-cycle arrest within the cell G₀/G₁ phase. Environmental and chemical growth inhibition results in a higher extent of cell-cycle arrest compared to the control conditions. While environmental growth inhibition resulted in constant cell-cycle arrest, cells seemed to adapt to the presence of the chemical growth inhibition, leading to a lower extent of cell-cycle arrest over time. © Reprinted from *Biotechnology Journal*, Moritz K. F. Wolf, Aurélie Closet, Monika Bzowska, Jean-Marc Bielser, Jonathan Souquet, Hervé Broly and Massimo Morbidelli, 'Improved performance in mammalian cell perfusion cultures by growth inhibition', 2019, with permission from John Wiley and Sons.

cell-cycle arrest, cells seemed to adapt to the presence of valeric acid when chemical growth inhibition was implemented. In particular, while an initial cell-cycle arrest was observed, after 7 days, cell proliferation started to increase again, as seen in Figure 4.16. Although the detailed mechanism of cell growth inhibition is not fully understood and it is expected to be highly cell-line dependent, these results demonstrate the potential of targeted growth inhibition during the production phase of a perfusion bioreactor.

A similar but more specific approach was chosen by Du et al. (2015) for the intensification of a non-steady-state perfusion process. Pyridopyrimidine-type molecules, which are able to arrest cells in the G₀/G₁ phase by target-specific binding to the cyclin-dependent kinases that controls the cell cycle, were used. By adding this molecule to the perfusion medium, the cell cycle was fully arrested (100 per cent) and cellular productivities were highly enhanced. For one cell line, the cell-specific productivity increased from 54 pg/cell/day to more than 110 pg/cell/day (Du et al. 2015). The applicability of such a molecule in a steady-state perfusion process needs to be verified, as such a process requires a minimum cellular growth in order to maintain a long-term stable culture.

Lin et al. (2017) and Wang et al. (2018) investigated different approaches to improve the cell culture media for steady-state perfusion application. In the first approach, the medium was reduced in salt components in order to further concentrate essential media ingredients and thus increase the media depth (Lin et al. 2017). In the second step, amino acid rebalancing enabled the reduction of lactate and ammonia levels and, therefore, the improvement of the cell-specific productivity without compromising viable cell density. In particular, these changes led to a cell-specific productivity increase from 40 to 80 pg/cell/day.

In a different study, Wang et al. (2018) tested the possibility of modulating and controlling cell-specific growth without adding any chemicals or supplements

4.5 Product Quality Attributes: Considerations on Their Control

by investigating the impact of extracellular concentrations of sodium (Na) and potassium (K) in cell growth and productivity. Absolute potassium concentrations are involved in the regulation of the membrane potential. Changes in the membrane potential are involved in cell-cycle progression, and elevated K levels are reported to induce cell-cycle arrest (Wang et al. 2018). By reducing the Na:K ratio from 9 (control medium) to less than 1 (modified medium), the cell growth was significantly suppressed. The higher K content resulted in induced cell-cycle arrest in the G0/G1 phase. Over the time course of a culture at 40×10^6 cells/mL, Wang et al. (2018) compared the control and the modified medium. While the control medium resulted in cell-specific bleed rates between 0.3 and 0.4 RV/day, leading to significant product loss and consistent cell-specific productivities of about 20 pg/cell/day, the modified medium led to decreased bleed rates of 0.1 RV/day and consistently increased cell-specific productivities up to 115 pg/cell/day. This study further supports the conclusion that targeted cell-growth arrest improves process yields by decreasing bleed volumes, but, can also lead to elevated cell specific productivities. The approach of Wang et al. (2018), in particular, is very powerful, as Na and K are standard media components and do not require further safety tests.

4.5 PRODUCT QUALITY ATTRIBUTES: CONSIDERATIONS ON THEIR CONTROL

A fundamental issue in the production of biotherapeutics are the quality attributes of the produced molecule, which is the subject of the last step of the culture development procedure illustrated in Figure 4.5. As discussed in Chapter 1, by product quality attributes, we mean, in particular, product-related impurities and substances – such as protein aggregates and fragments, charge and glyco isoforms – that are produced from the target product as a result of events like aggregation, oxidation, deamidation, isomerisation and others (Eon-Duval et al. 2012). Other quality attributes refer to process-derived impurities such as DNA, host cell protein, raw-material-derived impurities, protein A and others. All of these have to be kept within strict specifications. We have already seen that the narrower residence-time distribution inside a perfusion bioreactor, compared to a fed-batch one, is beneficial with respect to product quality. Here, we want to discuss how to control and modulate product quality attributes in a perfusion bioreactor through the continuous addition of appropriate supplements.

In the following sub-sections, we present different scenarios where considerations on product quality play a role.

4.5.1 New and Accepted Drug Molecules

When developing a new molecule, it is important to understand how critical process parameters affect the various product quality attributes, with the values corresponding to the molecule which undergoes the clinical trials providing the reference. The relevant issue is therefore that such quality profiles are reproduced at the larger commercial scale and then maintained unchanged and consistent during the entire process life time. This is particularly challenging in the case where the process conditions have to be changed, for example, to increase productivity in the case where an existing drug is found to be active for other indications sharing the same

genetic origin. In such cases, the product quality attributes have to remain within given specifications, independent of the changed process operating conditions.

4.5.2 Biosimilars

A different scenario refers to the production of a biosimilar. Here the product quality profile of the produced biosimilar has to reproduce that of the originator on the market within very narrow tolerance (Chugh & Roy 2014, Vulto & Jaquez 2017). It is important to understand how a robust manufacturing process can result in a highly similar molecule with consistent product quality. That is, glycosylation and charge variant patterns, aggregate and fragment levels and other critical attributes need to match the specifications of the originator.

4.5.3 Control of Specific Quality Attributes

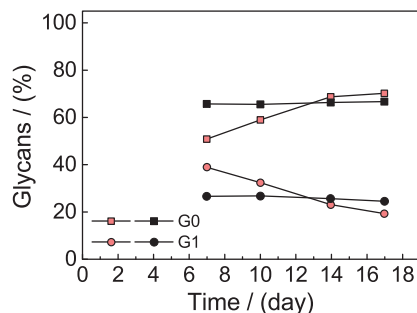
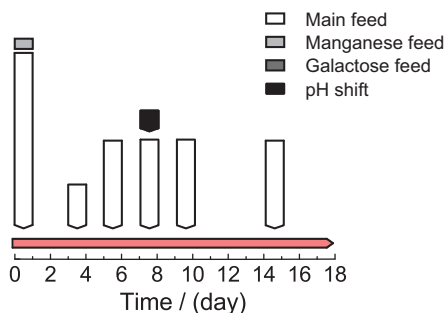
As downstream purification is capable of adjusting most of the critical attributes within certain boundaries but not the N-linked glycosylation pattern, this needs to be controlled in the upstream cell culture process. N-linked glycosylation, in fact, plays a crucial role for the therapeutic efficacy and safety of the protein and, therefore, needs to be carefully kept under control.

The N-linked glycosylation pattern can be modulated both in fed-batch and perfusion cell culture processes. Villiger et al. considered a fed-batch culture and developed an optimal strategy, based on extracellular pH control, and manganese and galactose additions, to maintain a constant glycosylation pattern during the entire process (Villiger et al., 2016a, 2016b). The obtained results are illustrated in Figure 4.17, where on the left-hand side the traditional and optimised feeding strategy are compared, with the arrows indicating the main feed addition (white), the addition of manganese (light grey), the addition of galactose (dark grey) and the adjustment of the extracellular pH (black). On the right-hand side, the percentage of the different glycoforms present in the reactor in the two cases are compared as a function of time. In the optimised feeding protocol, the manganese concentration was adjusted not only immediately after the inoculation but also on days 7, 10 and 14. Additionally, the feeding of galactose on days 7, 10 and 14 was also implemented. It is remarkable that the optimal feeding strategy was developed based on a predictive mathematical model, which is further discussed in Chapter 6. Nevertheless, glycosylation patterns represent only one out of several critical product quality attributes, and if others such as charge variant patterns and fragments also need to be maintained consistently, it may become extremely difficult.

Along the same lines, and based on a similar mathematical model, Karst et al. (2017b) extended this study to perfusion processes. On the one hand, it was shown that, differently from the fed-batch case considered before, during steady-state operation in a long-term perfusion, the same glycoform distribution was produced as a function of time. On the other hand, the possibility to modulate such glycosylation profiles by properly acting on the viable cell density and the media supplement feed rates (galactose and manganese) was investigated. A preliminary set of experiments was designed to investigate the effect of such quantities on the glycoform distribution. These were executed in a benchtop bioreactor, where different steady states were achieved sequentially within a single long-term perfusion run. It was found that high ammonia levels, obtained when increasing the viable cell density, inhibited the

4.5 Product Quality Attributes: Considerations on Their Control

Traditional strategy



Optimized strategy

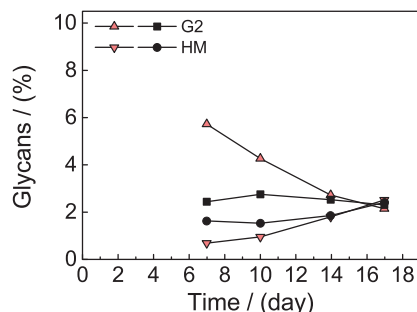
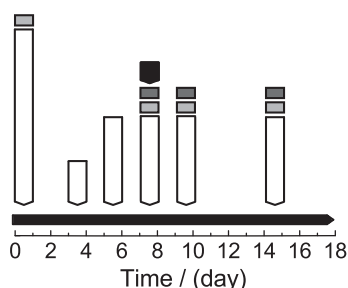


Figure 4.17 Traditional (red) and model based optimised feeding (black) strategy for a fed-batch process for N-linked glycosylation control (left). Comparison of the glycosylation pattern during the time course of the fed-batch process using the traditional (squares) and the optimised (circles) feeding strategy (right).

formation of complex galactosylated glycan structures, while the supplementation of either galactose or manganese, as well as their synergetic combination, significantly enhanced the formation of more complex glycan structures. This is illustrated by the three sequential steady states shown in Figure 4.18. These experiments were used to tune the parameters of the deterministic model, similar to the one mentioned earlier in the context of semi-batch operation, used to support the optimisation procedure. Indeed, the deterministic modelling approach combined with the sequential bioreactor runs enabled Karst et al. (2017b) to predict the system behaviour outside the reactor operating parameter space investigated experimentally, thus providing an efficient tool to develop continuous processes with controlled N-linked glycosylation patterns. This is a very interesting feature of mathematical models based on the physicochemical and biological processes occurring in a cell culture that is extensively discussed in Chapter 6.

In another case study, Bielser et al. (2019a) compared the impact of different reactor operating modes on a critical quality attribute of a conjugated protein. As shown in Figure 4.19A, this protein tends to be cut, or clipped, in two undesired pieces whose percentage represents an important quality attribute. In this work, three reactor operating modes are considered: low (LS) and high (HS) seeding fed-batch and perfusion (PF), as shown in Figure 4.19B. As expected, in both fed-batch applications, the protein accumulates in the system, and the titre increases in time. In perfusion, the protein is constantly harvested; the titre is, therefore, lower and

4 Design and Optimisation of Mammalian Cell Perfusion Cultures

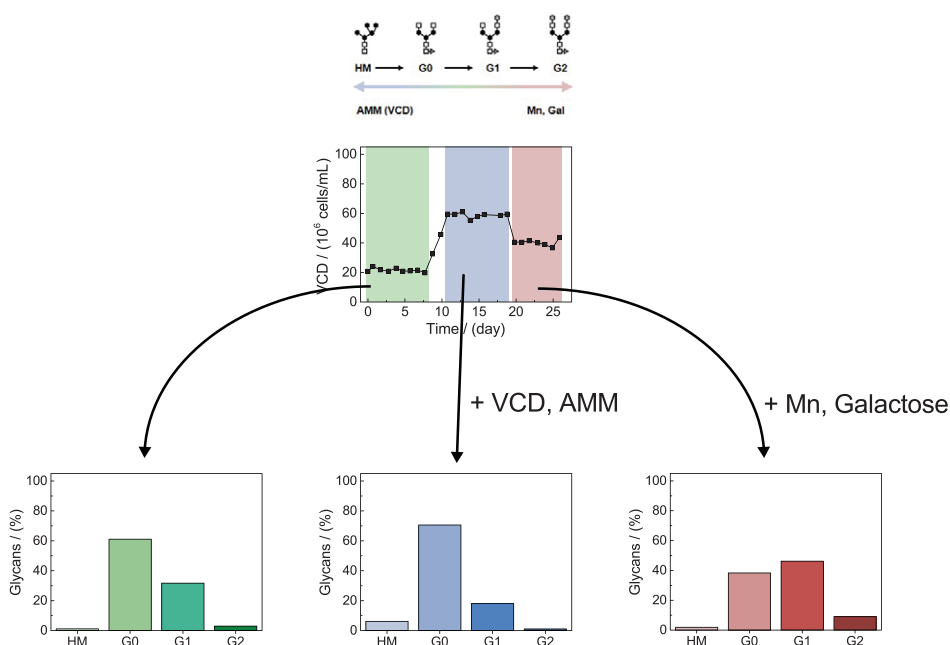


Figure 4.18 Results of the perfusion study of Karst et al. (2017b). Sequence of three steady states in a perfusion bioreactor at different VCD values: 20, 60 and 40 $\times 10^6$ cells/mL (center); in the third one, a supplement of galactose and manganese was added to the feed stream. It is seen that while the increase of VCD (centre in blue) from 20 to 60 $\times 10^6$ cells/mL leads to lower contents of complex glycans, the supplementation of galactose and manganese (right in red) favours more complex glycoforms. © Reprinted from *Biotechnology and Bioengineering*, Daniel J. Karst, Ernesto Scibona, Elisa Serra, Jean-Marc Bielser, Jonathan Souquet, Matthieu Stettler, Hervé Broly, Miroslav Soos, Massimo Morbidelli and Thomas K. Villiger, ‘Modulation and modeling of monoclonal antibody N-linked glycosylation in mammalian cell perfusion reactors’, 2017a, with permission from John Wiley and Sons

stable (Figure 4.19C). More importantly, the average residence time of the protein inside the reactor is smaller. As a result, the percentage of clipped molecules is larger in batch-type operations. In particular, in Figure 4.19D, this is seen to be 8.7 per cent and 4.9 per cent for LS and HS fed-batch, respectively, whereas it varied between 0.6 per cent and 1.5 per cent in the perfusion run. This highlights once more the potential benefit of continuous cell culture also regarding product quality, especially with the modern tendency of producing more and more complex molecules.

In summary, we have seen that perfusion bioreactors provide a very useful tool for the control of product quality and particularly the glycoform distribution, which is difficult to alter in downstream processes. Feed supplementation is very efficient in modulating glycan patterns in both fed-batch and perfusion processes. However, since perfusion allows more homogeneous or clean distributions (Karst et al. 2018, Walther et al. 2018) their modulation can be more accurate. In addition, the concept of operating the benchtop bioreactor across a sequence of steady states allows a more efficient exploration of different operating conditions – that is, one glycoform distribution per week compared to the two to three times longer times needed in the case of fed-batch operation.

4.6 Part 3: Scale-Up to clinical and Commercial Reactors

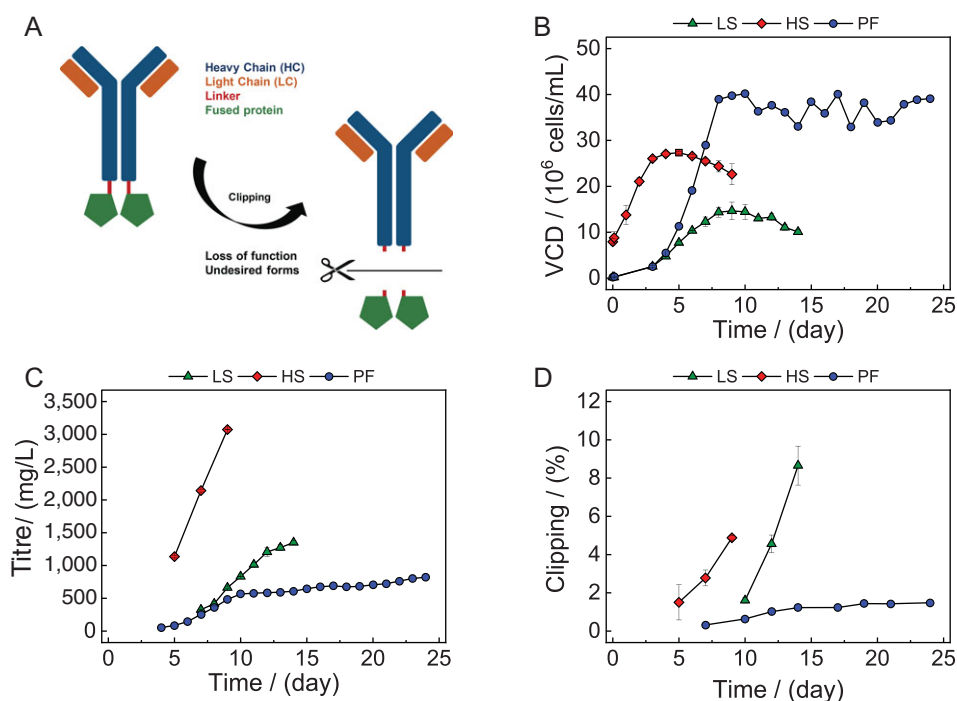


Figure 4.19 Example of a conjugated protein composed of a mAb and attached to another protein with a linker (A). Viable cell density (B), titre (C) and percentage of clipped forms (D) as a function of time for three different types of process: low (LS) and high (HS) seeding fed-batch and perfusion (PF). © Reprinted from *Journal of Biotechnology*, 302, Jean-Marc Bielser, Loïc Chappuis, Yashi Xiao, Jonathan Souquet, Hervé Broly and Massimo Morbidelli, 'Perfusion cell culture for the production of conjugated recombinant fusion proteins reduces clipping and quality heterogeneity compared to batch-mode processes', 26–31, 2019, with permission from Elsevier

4.6 PART 3: SCALE-UP TO CLINICAL AND COMMERCIAL REACTORS

Figure 4.20 schematically summarises the various steps necessary to scale to the commercial scale the perfusion culture that we have developed in the previous sections. The first one takes place at the mL scale or below, using reactor units like shake tubes, deepwell plates, and microscale bioreactors. Here, a large number of experiments is conducted with the objective of media and clone selection, but also of starting preliminary investigations of suitable process parameters for the perfusion process at the litre scale: usually the benchtop bioreactor. This is the scale where we optimise process performance, including productivity, yield and medium consumption, and make sure that all product quality parameters are within specifications. Based on our understanding of the effect of scale on processes like reactions, mass transport, mixing and others, it is, in fact, possible to conclude that this is the minimum scale for an accurate and reliable transfer of the process to the larger scales needed for clinical and commercial productions. This is largely based on the theory of multiphase reactors, which is well developed in the frame of chemical engineering and will be discussed in Chapter 5. The commercial scale required for perfusion reactors, even for expected large demands on the market such as for some monoclonal antibodies, is still relatively small, probably in the order of 500–2,000 L. These are significantly

4 Design and Optimisation of Mammalian Cell Perfusion Cultures

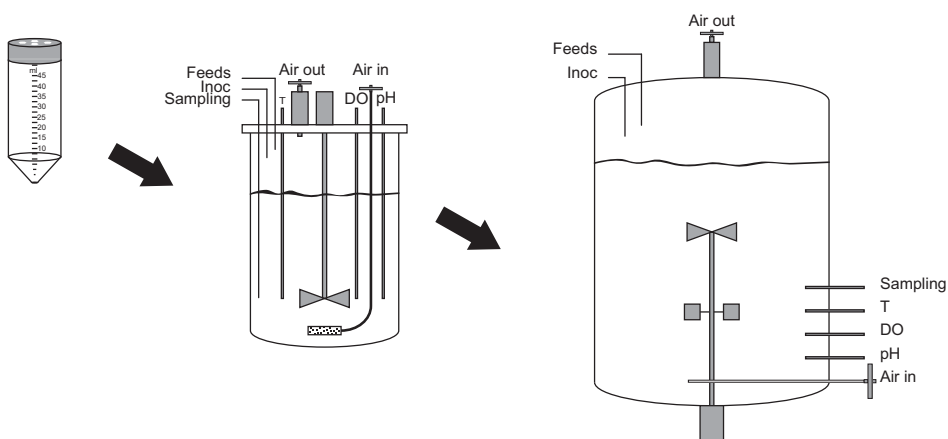


Figure 4.20 Schematic representation of the different reactor set-ups needed for the development of perfusion cultures at the commercial scale. These start from mL scale devices, such as shake tubes, move to the L scale of benchtop bioreactors and end up at the 1,000 L scale of the largest commercial units. © Reprinted from *Biochemical Engineering*, 131, Thomas K. Villiger, Benjamin Neunstoecklin, Daniel J. Karst, Eric Lucas, Matthieu Stettler, Hervé Broly, Massimo Morbidelli and Miroslav Soos, 'Experimental and CFD physical characterisation of animal cell bioreactors: From micro- to production scale', 2018, with permission from Elsevier

smaller than the counterpart fed-batch reactors that, due to their lower productivity, require larger volumes in the range of 10,000–20,000 L.

One relevant aspect, with respect to scale up of perfusion bioreactors, relates to the ancillary equipment and particularly to the cell retention device which is rather unique and specific for perfusion cultures. This involves an external loop and a filtering device, whose design and operation are scale sensitive. Mass transfer in the loop is generally not very efficient, so we need to make sure that the oxygen in the supernatant entering the external loop is not completely depleted before the loop exit. This requires sufficiently short residence times, which, however, need to be compatible with shear stress values that should not damage the cells.

4.7 CONCLUSION

In this chapter, we have presented the most important steps in order to develop and design a perfusion process for the commercial scale. This includes (1) the selection of a suitable combination of media and clones (in total, 1–2 media and 1–2 clones) with proper growth and productivity characteristics; (2) determining the $\text{CSPR}_{\min}$ for the given expression systems in a suitable scale-down system and (3) validation, optimisation and fine-tuning of the process parameters at the litre scale. This represents the suitable system for further scale-up to the commercial scale. Steps 1 and 2 are preferable performed at the nL to the mL scale, while step 3 needs to be verified at the L-scale. With respect to scale-up to the commercial scale, we introduce the critical scale-up parameters and the suitable design criteria in the next chapter. The important teaching of this chapter is how to take advantage of suitable small-scale models and reduce medium consumption and work effort by following a rational approach to

4.7 Conclusion

process development. Even if the ideal scale-down model for perfusion cultures is not yet available, the bench system is not always needed at the early stages of the design of a perfusion culture. Furthermore, ongoing technology advances will improve the current state-of-the-art scale-down models. It is only a question of when perfusion processes will be fully reproducible in the mL scale, as it is already possible for batch technologies.